Preserving the Fairness Guarantees of Classifiers in Changing Environments: A Survey

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The impact of automated decision-making systems on human lives is growing, emphasizing the need for these systems to be not only accurate but also fair. The field of algorithmic fairness has expanded significantly in the past decade, with most approaches assuming that training and testing data are drawn independently and identically from the same distribution. However, in practice, differences between the training and deployment environments exist, compromising both the performance and fairness of the decision-making algorithms in real-world scenarios. A new area of research has emerged to address how to maintain fairness guarantees in classification tasks when the data generation processes differ between the source (training) and target (testing) domains. The objective of this survey is to offer a comprehensive examination of fair classification under distribution shift by presenting a taxonomy of current approaches. The latter is formulated based on the available information from the target domain, distinguishing between adaptive methods, which adapt to the target environment based on available information, and robust methods, which make minimal assumptions about the target environment. Additionally, this study emphasizes alternative benchmarking methods, investigates the interconnection with related research fields, and identifies potential avenues for future research.

CCS Concepts: • **Computing methodologies** → **Learning paradigms**; **Learning under covariate shift;**

Additional Key Words and Phrases: Algorithmic fairness, trustworthy machine learning, distribution shift, covariate shift, uncertainty, online learning, distributionally robust optimisation

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1 INTRODUCTION

Machine Learning (ML) systems are increasingly being used for high-stake decision making in applications where human outcomes are involved, ranging from healthcare [84, 98], education [48], recruitment [26, 47], bank loan provision [38, 46, 59], to criminal incarceration [4, 20]. Several decision-making approaches have been found to reproduce social discriminating behaviours: The effects of their predictions are disproportionate for the different population subgroups described by sensitive attributes (e.g., gender, race, age). Hence, considering their significant societal implications, it is of utmost importance for these systems to exhibit fair performance. Addressing this concern has become a prominent focus within the research community, as reflected in recent studies [129]. Consequently, a wide range of fairness metrics and a substantial pool of methods have been developed to promote fair algorithmic behaviors across various ML tasks.

The evaluation of these models is mostly performed under single domain assumptions. That is, settings where the training and testing data are **independently and identically drawn (*i.i.d.*)** from the same distribution are usually considered. Nonetheless, the above assumption is quite strong and is not always warranted [137]. In practice, when these approaches are deployed in real-world settings, the deployment environments may and do systematically differ from the training environment. This phenomenon is known as *distribution shift* and might arise due to a variety of factors, such as, the non-stationarity of the data under study [29], which is very common in geographic and temporal data. For instance, the demographic composition of countries within the **European Union (EU)** has undergone substantial transformations in recent years, primarily driven by factors such as the refugee crisis and internal migration. These phenomena have led to notable shifts in the non-native population across various EU member states. Additionally, access to the data is not generally feasible from all possible test environments at training time, including the cases when the target distributions correspond to future situations. For example, in natural language processing applications [11, 12, 51, 135] the possible contexts in which the system may be deployed are infinite, and it is unfeasible to account for all of them at training time.

The dissimilarity between training and deployment environments can significantly deteriorate and potentially cause harm in fairness-critical applications [19, 29, 41]. In the cases where distribution shift occurs, fairness requirements are satisfied on training samples, while they may be severely violated in the deployment data [55, 64]. For instance, in practical healthcare applications, a disease prediction model that demonstrates fairness guarantees based on specific fairness criteria within one hospital setting may exhibit unfair behavior when deployed in a hospital serving a different community. In the light of this new potential source of unfairness, accounting for possible distribution shifts in the context of fair machine learning is crucial: Providing robust fairness guarantees under conditions of distributional shift is necessary for the safe deployment of fair decision-making models.

In recent times, numerous efforts have been made to address these challenges, resulting in a notable growth in the field. However, there remains a lack of a comprehensive taxonomy for fairness-aware methods in changing environments, making it difficult to compare different approaches. Through this survey, we strive to establish a unifying framework and offer a comprehensive overview of this burgeoning research area. Our approach includes taxonomy analysis,

comparison, benchmarking proposals, and connections to related research fields, providing valuable insights into the advancements in this domain.

The reminder of this article is organised as follows. Section 2 briefly introduces the reader into the theoretical framework this survey is grounded in, specifying the notation, the main definitions, and stating the motivating problem. A clear classification of the different kinds of distribution shifts is also presented. Then, Section 3 highlights the parallelisms between the field of distribution shift and fairness and discusses whether specific approaches that account for both are necessary. Our main contribution is provided in Section 4, which is a taxonomy-based overview of the research field. Section 5 describes the existing datasets and benchmarking frameworks, and Section 6 draws connections with other related research areas. In Section 7, we discuss the interest of the present survey, noticing some of the current real-world applications and highlighting the future venues on the field. Finally, in the Appendix, Table 4 provides a summary of the main characteristics of the studied approaches to preserve fairness guarantees of classifiers in changing environments.

2 THEORETICAL BACKGROUND

2.1 Notation

This section describes the notation adopted throughout this survey. A domain $\mathcal{D}$ is characterised by a feature space $\mathcal{X} \subseteq \mathbb{R}^d$, a sensitive attribute space $\mathcal{A}$, an output or label space $\mathcal{Y}$ and an associated joint probability distribution $\mathbb{P}(x, a, y)$ over the feature-label space triplet $\mathcal{X} \times \mathcal{A} \times \mathcal{Y}$; that is, $\mathcal{D} = \{\mathcal{X}, \mathcal{A}, \mathcal{Y}, \mathbb{P}(x, a, y)\}$. Each instance is described by three random variables: $X \in \mathcal{X}$ non-sensitive features, $A \in \mathcal{A}$ the sensitive attribute and $Y \in \mathcal{Y}$ the class label. The sensitive attribute is usually related to demographic characteristics, including gender, age, or race, that divide the population into distinct subgroups. Among these subgroups, there is often a prominently favoured group that aligns with the majority, as well as one or more potentially disadvantaged groups that represent minority or marginalized subgroups. In formal terms, the sensitive attribute space, denoted as $\mathcal{A}$, represents the set of possible values for the sensitive variable. This space can be discrete, such as in the case of binary ($\mathcal{A} = \{0, 1\}$) or multivalued ($\mathcal{A} = \{1, \ldots, K\}$) attributes like gender or race, or it can be continuous ($\mathcal{A} \subset \mathbb{R}$) as seen with attributes like age. In the general multidimensional scenario, the sensitive attribute space is constructed as the product of the individual supports of each sensitive variable.

In particular, we consider two domains: a source domain $\mathcal{D}_s$ and a target domain $\mathcal{D}_t$. The source domain will represent the population in which the original classifier has been trained and the target domain the population of the environment in which the classifier is about to be deployed. In practice, we assume to have access to a set of n instances from the source domain defined by feature-label triplets $\{(x_i, a_i, y_i)\}_{i=1}^n$ *i.i.d.* from $\mathbb{P}_s(x, a, y)$. With regards to the target domain, we can either have access to m labelled samples with or without the sensitive information, namely $\{(x_i, a_i, y_i)\}_{i=1}^m$ or $\{(x_i, y_i)\}_{i=1}^m$, m unlabelled samples with or without the sensitive information, $\{(x_i, a_i)\}_{i=1}^m$ and $\{x_i\}_{i=1}^m$, respectively, or we might even have no information at all. The available target samples are assumed to be *i.i.d.* from $\mathbb{P}_t(x, a, y)$.

The classifiers analyzed in the survey aim to identify the optimal parameter setting $\theta \in \Theta$ for the mapping function $f : \mathcal{X} \times \mathcal{A} \longrightarrow \mathcal{Y}$ that will most accurately and fairly describe the real data generating process of the deployment (target) environment. Since most of the literature in the field of algorithmic fairness adopts a binary classification task, we will stick to this framework for sake of clarity. The optimisation problem will be defined by a suitable loss function $\mathcal{L}(\theta, X, A, Y)$ and in some cases with an added set of constraints $\boldsymbol{g}(\theta, X, A, Y)$. Furthermore, we will use the notation p specifically for representing probabilities of individual events (should not be confused with $\mathbb{P}$, which describes the probabilities of various events within probability distributions).

Table 1. Notation Adopted in This Survey

Notation	Description
(x, a)	Instance
y	Label
$\mathcal{X}, \mathcal{A}, \mathcal{Y}$	Feature/Sensitive feature/Label space
$\mathcal{D}$	Domain
$\mathbb{P}(\cdot)$	Probability distribution
$\mathbb{E}[\cdot]$	Expectation
$\mathcal{L}(\cdot)$	Loss function
$\boldsymbol{g}(\cdot)$	Constraints
p	Probability of an individual event
$f(\cdot)$	Predictive function
θ	Model parameter
Θ	Parameter space
s, t	Subscripts denoting the source/target domain
n, m	Dataset size of the source/target domain

The described notation is summed up in Table 1.

2.2 Algorithmic Fairness

The Article 21 of the EU Charter of Fundamental Rights dictates that "any discrimination based on any ground such as sex, race, colour, ethnic or social origin, genetic features, language, religion or belief, political or any other opinion, membership of a national minority, property, birth, disability, age, or sexual orientation shall be prohibited." It describes two different discrimination scenarios: (1) *direct discrimination* (disparate treatment in the U.S. law): intentionally providing an adverse treatment based on the sensitive attribute; and (2) *indirect discrimination* (disparate impact): when a seemingly "neutral provision, criterion or practice" disproportionately disadvantages members of a given sensitive group compared to others. This last type of discrimination highlights the systematic and structural unfairness found in the society. Nonetheless, it cannot be measured at individual level, since it requires a comparison between different groups [110]. Algorithmic fairness pertains to the concept and field of study focused on designing and implementing algorithms in a way that avoids bias and promotes equitable outcomes across different groups of individuals. It aims to ensure that the decisions made by algorithms are not discriminatory or unjustly favour or disadvantage certain individuals or groups based on sensitive attributes such as race, gender, age, or socioeconomic status.

2.2.1 Measuring Algorithmic Fairness. A long list of definitions have been proposed to measure the unfair behaviour of algorithms (refer to Reference [109] for an extensive review), which are closely related to legal notions for discrimination. Fairness metrics can be broadly categorized into three groups: individual notions, group (or statistical) criteria, and mini-max fairness. Individual notions focus on ensuring that similar individuals are treated similarly by the algorithm. Statistical definitions aim to achieve approximate parity of one or more performance metrics across different groups. Eventually, mini-max notions of fairness aim to enhance the predictive performance on the sensitive group with the poorest recorded performance.

Formally, the notion of *individual fairness* [32] defines a similarity metric d with respect to the particular context under consideration, and requires that instances that are similar according to such metric receive similar treatment (i.e., outcome) by the algorithm:

$$|p(\hat{y}_i = y|x_i, a_i) - p(\hat{y}_j = y|x_j, a_j)| \leq \epsilon \quad \text{when } d((x_i, a_i), (x_j, a_j)) \approx 0. \tag{1}$$

The main drawback of individual notions is that they are subject to many strong assumptions, such as the knowledge about the structure of the similarity metric, which is actually non-trivial. Furthermore, generating all possible data for testing this criterion is computationally infeasible.

However, there are *statistical* or *group notions* of fairness, which are the most common criteria in the literature regarding algorithmic fairness, since they are easier to measure and implement. Group notions of fairness study the gap of one (or multiple) performance metric(s) between different sensitive groups. The most popular statistical definitions are *demographic parity*, *equality of opportunity* and *equalized odds*.

***Demographic parity* (DP)**, also known as ***statistical parity* (SP)**, requires that the probability of having a positive outcome be similar across sensitive groups, and is closely related to the legal definition of *indirect discrimination*. In other words, this notion claims that among the instances having the desired and undesired outcomes, the proportion of people with given sensitive attribute should be the same and therefore, match the demographic statistics across the population as a whole:

$$|p(\hat{Y} = 1|A = a) - p(\hat{Y} = 1|A = a')| \leq \epsilon. \tag{2}$$

This fairness definition is troublesome in what models might assign positive outcomes to the instances of the unprivileged group at random to satisfy the criterion, without any guarantees on whether those that really deserve it are being considered. Besides, it is also controversial in situations where the base rate (i.e., prevalence) significantly differ among different sensitive groups.

To overcome such issue, several researchers proposed different criteria that account for the true class label in their definition, assuming that class labels are free from biases. ***Equality of opportunity* (EOp)** requires the ***true-positive rate* (TPR)** to be similar across sensitive groups. In other words, it requires that the probability of having a positive outcome when the true label is positive to be similar across different subgroups:

$$|p(\hat{Y} = 1|A = a, Y = 1) - p(\hat{Y} = 1|A = a', Y = 1)| \leq \epsilon. \tag{3}$$

Further, ***equalized odds* (EO)** requires approximate parity not only on the TPR but also on the ***false-positive rates* (FPR)**:

$$|p(\hat{Y} = 1|A = a, Y = y) - p(\hat{Y} = 1|A = a', Y = y)| \leq \epsilon \quad \forall y. \tag{4}$$

In many applications, enforcing statistical notions of fairness has been found to compromise the predictive accuracy of the classifier [21, 65, 75]. Nonetheless, sacrificing accuracy is often unacceptable, specially in critical settings, such as, healthcare, where the no-harm principle [107] must be satisfied. In such scenarios, models are built with the aim of achieving good performance across different predefined sensitive groups [27]: for that purpose, they are trained to minimise the error of the worst group (i.e., minimise the error of the group with the highest error), as originally done by Rawls in References [89, 90]. This approach gives rise to the notion of fairness known as mini-max fairness.

2.2.2 Fairness-enhancing Methods. The research area of algorithmic fairness has come up with a massive pool of alternatives to enforce fairness in different ML tasks [83]. Those approaches are usually grouped into three categories subject to the point in which the fairness-enhancing intervention is performed throughout the task: they are classified into pre-processing, in-processing, and post-processing methods. *Pre-processing* methods [14, 34, 58, 67, 88, 124] transform the data to obtain an unbiased dataset to feed a standard ML classifier, whose output is expected to be fair. Preliminary works (e.g., Reference [56]) proposed to change the labels of some instances (massaging) or reweighing them to obtain fair datasets. Nonetheless, the latest approaches bring forward the idea of learning a new and fair representation of the data [124]. *In-processing* methods

[2, 7, 13, 42, 57, 87, 118, 121, 122] modify the algorithm itself to fulfil fairness criteria at training time. The latter is usually done by adding a regularisation term to the loss function that penalises unfairness (e.g., Reference [57]) or by considering additional constraints on the optimisation problem that account for fairness requirements (e.g., References [118, 122]). *Post-processing* approaches [21, 33, 46, 75] adjust the output of a classifier so that the final performance is fair. For example, in Reference [46] the authors suggest a technique for flipping some of the classifier's predictions so that the final decisions satisfy fairness requirements.

Several Python packages, including `AIF360` [9], `Fairlearn` [115], and `Themis-ML` [6], offer comprehensive toolkits and libraries for addressing algorithmic fairness, providing a wide range of fairness metrics and bias mitigation techniques. In addition to the aforementioned Python packages, there are also several R packages available for addressing algorithmic fairness, including `fairness` [62], `Aequitas` [94], `fairml` [97], and `fairmodels` [117].

2.3 Changing Environments

Most of the fairness-enhancing interventions assume that the ML algorithm will be deployed in a population that is identical to the one the classifier has been trained on. However, such ideal conditions are hardly held in real-life applications. In fact, there usually exists a *distribution shift* between the training and target data, i.e., $\mathbb{P}_s(x, a, y) \neq \mathbb{P}_t(x, a, y)$. This can happen either because the classifier will be deployed in a different population, or because the source population itself is evolving.

Consider a joint probability distribution consisting of three factors U, V and W; this joint probability can be decomposed as follows:

$$\mathbb{P}(U, V, W) = \mathbb{P}(W|U, V)\mathbb{P}(V|U)\mathbb{P}(U). \tag{5}$$

The factors are defined as: $U \subseteq \{X, A, Y\}$, $V \subseteq \{X, A, Y\}\backslash U$, and $W \subseteq \{X, A, Y\}\backslash(U \cup V)$. A shift in the distribution can be caused by several different factors of this decomposition, which characterise the type of distribution shift. First, we can have changes in the marginal distribution of the features or the class label and, second, changes in the conditional distributions. Depending on which is the marginal whose distribution changes, the shift is named accordingly. Whereas, all the changes on the conditional distributions are named *concept shift*. That is, we may have the following distribution shifts:

- If the shift arises in the marginal distribution of some attribute, that is, if there exists $U \neq \emptyset$ for which $\mathbb{P}_s(U) \neq \mathbb{P}_t(U)$, then the distribution shift is named according to the characterisation of U:
  - ($U = A$) *Demographic shift* [10] arises when the distribution of the population w.r.t. the sensitive attribute varies, i.e., $\mathbb{P}_s(A) \neq \mathbb{P}_t(A)$. This shift emerges when e.g., the proportion of female individuals varies from the source to the target population.
  - ($U = X$) *Covariate shift* [92] is a special scenario in which the source and target domains have different distributions with respect to the features $\mathbb{P}_s(X) \neq \mathbb{P}_t(X)$, but the functional form remains unaltered across domains, i.e., $\mathbb{P}_s(Y|X, A) = \mathbb{P}_t(Y|X, A)$. For example, when, due to economic trends, the socioeconomic situation of a given population evolves over time, there is covariate shift between the distributions of the populations of subsequent years.
  - ($U = Y$) *Label shift* (or *prior probability shift*) [24] comes off when the distribution of the class label Y (also known as prevalence) varies from the source to the target domain, i.e., $\mathbb{P}_s(Y) \neq \mathbb{P}_t(Y)$. Furthermore, label shift also assumes that the distribution of the features conditioned on the class label and sensitive attribute is preserved across domains, i.e.,

$\mathbb{P}_s(X|Y,A) = \mathbb{P}_t(X|Y,A)$. Label shift pops up when, e.g., the prevalence of a given disease varies for different geographical regions.

  - U can also refer to any combination between X, A, and Y. In those cases, the shift is denoted as *compound shift*. For instance, we can have a compound of covariate shift and demographic shift, where $U = \{X, A\}$, $V = Y$ and $W = \emptyset$. $\mathbb{P}(V|U)$ will represent the functional form $\mathbb{P}(Y|X,A)$, which is assumed to remain unaltered across domains. Furthermore, a *general distribution shift* scenario arises if $U = \{X, A, Y\}$ and, therefore, $W, V = \emptyset$. In fact, the majority of the shifts that arise in real world settings are compound shifts.

— If there exists $U \neq \emptyset$ for which $\mathbb{P}_s(U) = \mathbb{P}_t(U)$, and at least $V \neq \emptyset$ for which $\mathbb{P}_s(V|U) \neq \mathbb{P}_t(V|U)$ or $W \neq \emptyset$ for which $\mathbb{P}_s(W|V,U) \neq \mathbb{P}_t(W|U,V)$, then it is a case of *concept shift*. Originally concept shift referred to the situation in which a changing context induces changes in the target concepts, i.e., when $\mathbb{P}_s(Y|X,A) \neq \mathbb{P}_t(Y|X,A)$. However, it was later adopted to denote changes in other conditional distributions [77]. Concept shifts are very common in online learning scenarios, particularly, the case where $\mathbb{P}(Y|X,A)$ varies. For example, in a recommender system, a user's interests may fluctuate due to a variety of reasons, such as, personal needs, current trends, employment status and age, which are not captured by the model, while the attributes that are actually accounted for ($\mathbb{P}(X,A)$ or $\mathbb{P}(X)$) remain unaltered. Furthermore, in healthcare applications, the progression of sick patients might vary in response to the applied medication or variations in the natural resistance of the patients.

3 SIMILARITIES BETWEEN DISTRIBUTION SHIFT AND FAIRNESS

In recent years, both distribution shift and fairness have been extensively investigated in the field of machine learning, leading to the development of closely related methods to tackle each of these issues. For instance, the literature of both research fields contains a relevant amount of works that focus on learning a new representation of the data that is invariant to a given change in the context. On the one hand, methods regarding distribution shift aim to learn representations that are invariant across domains, while in fairness the invariance property of the new representation is defined w.r.t. the sensitive information. Besides, methods that were explicitly developed to overcome distribution shift have been found to perform well to address fairness-related concerns [1, 113, 120], and vice-versa [23, 108], showing that both research areas share similar general characteristics. On the other hand, the picture is not so clear when both challenges appear at the same time: Do we need specific methods to address distribution shift in fairness-aware scenarios?

The naïve approach to tackle this problem is to employ conventional distribution shift methods in combination with fairness-enhancing methods, addressing each issue independently. In principle, the objective of achieving fairness can be pursued through both pre- and post-processing fairness-enhancing methods, as these approaches have the flexibility to be employed alongside any ML classifier. In a shifted scenario, post-processing methods could modify the predictions of any popular classifier that resolves distribution shift to make them satisfy fairness requirements (e.g., as it is done in Reference [55]). In a similar manner, pre-processing methods would modify a dataset so that fairness requirements are satisfied and then, feed it to a conventional classifier that accounts for distribution shift (e.g., Reference [49]). However, addressing the concerns related to fairness and distribution shift in a disjoint fashion might lead to sub-optimal solutions [3]. Therefore, it is preferable to tackle them jointly to preserve fairness guarantees and high predictive performance in the shifted environment, by modifying the conventional pre-, in- and post-processing fairness-enhancing approaches drawing inspiration from conventional shift-aware classifiers (or vice versa). For instance, many pre-processing methods aim to learn a new representation of the data that will be fair for the different sensitive groups: To account for distribution shift, such methods could be modified to make the new representation also be invariant across domains (e.g.,

Reference [96]). Besides, when it comes to in-processing methods, the optimisation problems for training the classifier need to re-defined so that they also account for distribution shift, to build models that are fair and accurate in the shifted deployment environment.

The literature in the intersection between enforcing fairness and preserving predictive performance under distribution shift has also studied the existence of possible trade-offs between both objectives, and several works have claimed that they might not be completely orthogonal. Building on that, recent research has explored under which conditions enforcing fairness implies robustness and vice versa. Madras et al. [68] empirically demonstrated that when fairness is adversarially learned following their method, the resulting representation is not only fair but can also transfer predictive performance as well as fairness guarantees to new tasks. Maity et al. [69] showed that enforcing group notions of fairness mitigates algorithmic biases caused by sub-population shifts (e.g., under-representation of a given group) whenever imposing such constraint recovers the Bayes' classifier in the target domain. Mukherjee et al. [79] studied the similarity between enforcing **individual fairness (IF)** and domain adaptation and generalisation approaches, and derived the conditions under which enforcing IF achieves domain generalisation and vice-versa. Although the works above address specific scenarios where enforcing one criterion provides safe guarantees on the other, the scenarios they describe are too demanding and, therefore, hardly satisfied in real-world applications. In the light of such issue, their utility to preserve fairness guarantees and predictive performance in shifted environments is limited. Consequently, their existence does not call into question the necessity to develop methods to jointly address fairness concerns and distribution shift under more realistic conditions.

4 OVERCOMING DISTRIBUTION SHIFT IN BIAS-AWARE SCENARIOS

In this section, we present the existing approaches that address robustness in performance while guaranteeing fairness in distribution shift frameworks. Depending on the information available about the deployment environment, methods that overcome the adversities produced by distribution shift have been developed in two different scenarios:

(A) If there is some information available from the target environment, then the classifier initially trained on the source environment is **adapted** to the target. This refers to the scenario where, for instance, Hospital A has an automated ML system designed to predict the presence of a specific disease in individuals and there is a need to adapt and deploy this system safely in Hospital B, which serves a different population.

(B) Alternatively, in the absence of information regarding the target environment, methods are trained to be **robust** against any potential deployment distribution. This situation arises, for instance, when the disease prediction tool developed in Hospital A aims to be safely deployed in any possible hospital.

Based on this distinction, the methods will be first classified into two main groups: **adaptive** methods and **robust** methods. The taxonomy proposed to characterize the different methods of this research field is summarized in Figure 1.

In the (A) cases in Figure 1, the kind of information known about the target environment is generally classified into three different types: data from the target population, the distribution from the target domain or the causal graph that jointly describes the data-generating processes from the source and target environments. The former situation gives rise to (A.1) **data-based** fair adaptive approaches and happens, e.g., when patient data is available in the target hospital. The second type considers the scenario (A.2) in which there is certain knowledge regarding the **probability distribution** of the target environment, e.g., we (partially) know the distribution that describes the population of the target hospital. Finally, the third type requires (A.3) **causal** fair adaptive

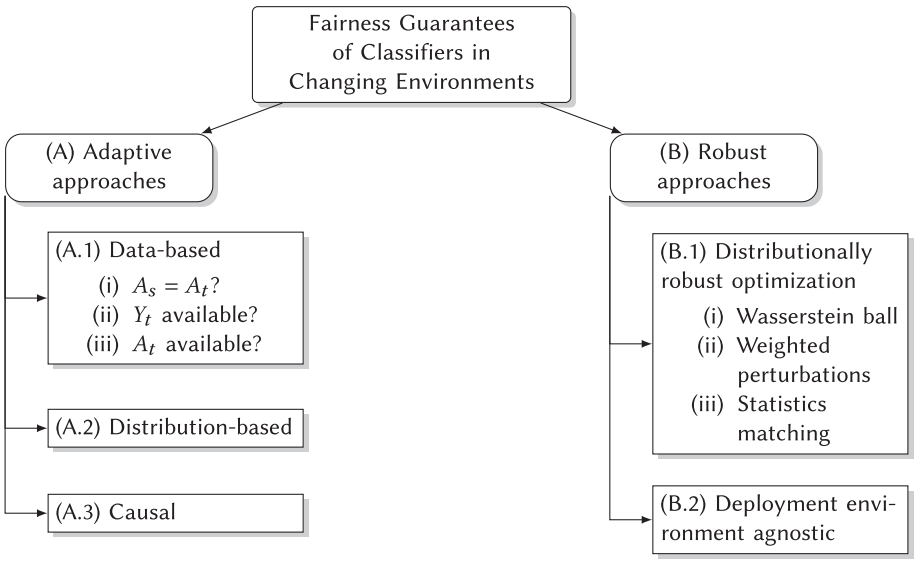


Fig. 1. A taxonomy of methods that address distribution shift challenges while enforcing fairness constraints. The approaches are categorized based on the availability of information from the target environment, distinguishing between **adaptive** methods, which adapt to the target environment based on available information, and **robust** methods, which make minimal assumptions about the target environment. Within the adaptive methods, further categorization is made according to the type of information available from the deployment environment, including **data-based**, **distribution-based**, and **causal** approaches. Within the data-based category, methods are further organized based on the characteristics of the available target information. Conversely, the fair robust approaches consist of fair **Distributionally Robust Optimization (DRO)** and **deployment environment agnostic** methods. DRO approaches are categorized based on their modeling of the distributional uncertainty.

approaches, and can be found e.g., when accessing to instances from different hospitals is out of hand, but we instead have knowledge about which feature's distributions are expected to vary across different hospitals and which of them are expected to remain unaltered.

However, in the context of robust approaches (B) in Figure 1, we will distinguish between methods that make assumptions about the deployment environment, specifically **Distributionally Robust Optimization (DRO)** models (B.1), and those that undergo a process of robustification by being agnostic to the deployment environment (B.2).

4.1 Fair Adaptive Approaches

The main objective of fair adaptive approaches is to adapt an existing classifier, that has been trained on a particular distribution, to ensure its safe deployment in an environment where the underlying data generation process may differ from the original distribution. This procedure is highly beneficial in three different scenarios. First, (A.1) when the target data is incomplete to build a new model from scratch either because there are few labelled examples, all instances are unlabelled or the sensitive information is lacking in the target environment. Further, it is as well an advantageous procedure when instances from the target environment are not available but some information is known either on the distribution (A.2) or on the causal graph (A.3).

Concerning the data-based methods (A.1), various scenarios can arise based on the assumptions made about the availability of the class label and sensitive information of the target data. It is important to note that, in every scenario, at least the values of the non-sensitive features X are

Table 2. Characterisation of the Possible Scenarios That Data-based Fair Adaptive Approaches May Face Regarding the Availability of the Class Label and the Sensitive Information in Target Data and the Assumptions Made on the Sensitive Attribute

Scenario	Target information		
	$A_s = A_t$	Y **available**	A_t **available**
1	✓	✓	✓
2	✓	✓	✗
3	✓	✗	✓
4	✓	✗	✗
5	✗	✓	✓
6	✗	✓	✗
7	✗	✗	✓
8	✗	✗	✗

known for the instances in the target environment. Each scenario entails a distinct data-based adaptation procedure, which can be characterized by three distinct orthogonal aspects:

(I) **Is the sensitive feature assumed to be the same across domains ($A_s = A_t$)?** The sensitive information might vary from one domain to another. For instance, in the source domain (e.g., Hospital A), the sensitive attribute might be gender, whereas in the target domain (e.g., Hospital B), it could be race. In such cases, classifiers are designed with the objective of meeting fairness criteria concerning gender in the source domain and race in the target domain.

(II) **Are labels available on target data?** In the cases where the target data is available, the instances may consist of labelled (supervised adaptation) or unlabelled examples (unsupervised adaptation). For instance, the available data pertaining to patients in the target hospital may or may not contain information regarding their actual disease status.

(III) **Is the sensitive information available in the target domain?** Similar to the previous consideration, the available data might lack from sensitive information (A_t). Indeed, the sensitive attribute of the patients of the target hospital (e.g., gender, race, age) might not be available.

Therefore, data-based fair adaptive approaches might face $2^3 = 8$ possible scenarios, that are detailed in Table 2, regarding the assumptions made on the class labels and sensitive information.

4.1.1 Data-based Fair Adaptive Approaches (A.1). These methods assume that some target data is accessible at training time and use it in combination to the labelled instances from the source domain in the training process. However, the datapoints from the source and target environments play different roles in the adaptation procedure, due to its distinct characteristics and information availability. Most of the methods are based on existing popular approaches that address distribution shift in a fairness-agnostic scenario, modified to account for fairness guarantees in shifted scenarios (besides accurate performance). Depending on the information available from the target environment and the assumptions regarding the sensitive information, eight different scenarios are described in Table 2. Note that, the more assumptions the methods make regarding the available information from the target, the less flexible they become. Consequently, the most flexible adaptive method would be the one that grounds on scenario 8. However, methods that do not assume $A_s = A_t$, can always be employed in the equivalent cases where $A_s = A_t$ holds (e.g., methods that address scenario 5 can be used to address scenario 1).

Among the existing data-based methods, the majority consider either scenarios 1 or 3, while there are even some that have not yet been covered by any method, specifically scenarios 2, 6, and 8. We acknowledge that scenarios 6 and 8 might require additional structure as they assume the presence of a different sensitive attribute in the target domain ($A_s \neq A_t$), yet information about the sensitive attribute in the target environment (A_t) is unavailable. More precisely, most of the works assume there is no access to the target label (unsupervised adaptive methods), and among those that do assume its availability (supervised adaptive methods) the majority are grounded on the online learning setting where the data comes in a streaming fashion. Furthermore, the sensitive information is more often than not assumed to be available: indeed very few approaches contemplate such information might not be accessible in the target domain (e.g., Reference [22]). Besides, the most popular assumption regarding the sensitive attribute is that it remains unchanged for the different domains, and therefore, those methods that assume that it may differ across domains are scarce (e.g., References [96, 119]). In what follows, we outline the various approaches that have been proposed to tackle each particular scenario. The scenarios are presented in descending order of complexity, starting with the ones that are relatively easier to address and progressing towards the more challenging ones (i.e., with fewer assumptions). Specifically, we commence with the scenarios that assume $A_s = A_t$.

(Scenario 1) Supervised adaptation with available sensitive information and $A_s = A_t$. This scenario is most commonly addressed in *online learning* settings, where the data is generated in a streaming fashion and its characteristics may evolve over time. Although it consists of a fully supervised setting, the label of an instance becomes available shortly after its arrival. In this framework, a prediction is made for every arriving instance using the current classifier, and the latter is updated after the ground truth labels of those instances become available. This setup is popularly named first-test-then-train or prequential evaluation. Iosifidis et al. [50] proposed a pre-processing procedure: if the unfair behaviour of a classifier exceeds a threshold value ϵ for the set of instances arriving at time t, the chunk is modified either by massaging or by re-weighing the instances before feeding it to the base learner. The base learner can be any classifier that operates in an online learning setting. Post-processing methods can also be employed in conjunction with classical online learning classifiers. Iosifidis and Ntoutsi [49] proposed one such method, applicable to decision boundary-based online learning classifiers. They keep track of the cumulative fairness guarantees of the classifier, to avoid long-term discriminatory effects that can go unnoticed when fairness concerns are taken into account for each particular time point. Whenever the cumulative discriminatory behaviour surpasses a used-defined threshold ϵ the decision boundary is adjusted to fulfil the fairness requirements.

Besides, there are several in-processing approaches that adopt a *decision-tree*-based approach [125–127]. In-processing approaches that operate in an online learning setting adopt two different strategies w.r.t. shift detection and the consequent model adaptation: In all the cases, when distribution shift is detected the model is adapted to the new instances subject to fairness requirements; however, some approaches assume that the distribution has shifted only when a deterioration in the predictive performance is detected (e.g., References [125, 127]), while others account not only for performance deterioration but also for rising fairness violations for shift detection [126]. For instance, among the first group of methods, Zhang and Bifet [125] proposed a new fairness-aware splitting criterion that reckons information and fairness gain. This criterion was further extended in Reference [127] by adding a parameter to the definition to control the trade-off between information gain and fairness gain: at each time point, the practitioner can interpolate between a model that favours accuracy and a model that favours fairness. However, Zhang et al. [126] brought forward a fair and adaptive **random forest (RF)** that belongs to the second kind of approaches: that is, changes in the distribution are not only characterised in terms of performance deterioration but

also with increasing unfairness. Besides, fairness guarantees are taken into consideration at various stages throughout the training and adaptation process: (i) base learners (i.e., decision trees) are chosen to increase ensemble diversity regarding discrimination, where each of them embodies a distinct representation of discrimination; (2) they under-sample the instances from the privileged group that have the positive (desired) class label to build fair base learners; and (3) the predictions of the base learners are weighted according to their current fairness guarantees.

Concerning the fairness notions considered in works addressing this scenario, while the initial approach by Iosifidis et al. [50] focuses on the conventional SP fairness concept, subsequent approaches [125–127] consider cumulative versions of SP. Furthermore, Iosifidis and Ntoutsi [49] incorporate a broader range of notions, including cumulative SP, cumulative EOp, and cumulative predictive equality [109].

(Scenario 3) Unsupervised adaptation with available sensitive information and $A_s = A_t$. Most of the data-based adaptive works consider this scenario: they assume that target data is unlabelled but that there is access to the sensitive feature. Further, they suppose that the sensitive feature is that of the source domain. The first works that addressed this scenario adopted a reweighing approach [22, 55]: they assign weights to the instances of the source domain to represent the population of the target domain, since the class labels are available only for the source data. Once these weights are known, it is possible to evaluate the expected value of any function on the target population using source samples. Depending on the fairness-enhancing strategy under consideration, the weighing function is employed and learned differently. For example, Kallus and Zhou [55] proposed a weighing scheme to be employed with the popular post-processing approach by Reference [46], which enforces fairness in terms of EOp and EO on the target data. Such fairness-enhancing method takes as input the *TPR* and/or *FPR* of a conventional ML classifier for the different sensitive groups, evaluated on the target domain. Since the labels of the target instances are unknown, the *TPR* and *FPR* are estimated using the source data, by weighting the contribution of each instance by the classical covariate shift instance weighing function, treating the sensitive attribute as a conventional feature:

$$w(x, a) = \frac{\mathbb{P}_t(x, a)}{\mathbb{P}_s(x, a)}. \tag{6}$$

Hence, in this method, the adaptation and the fairness-enhancing intervention are disjoint. However, Coston et al. [22] proposed a different framework considering an in-processing fairness-enhancing procedure, where fairness and distribution shift are jointly considered at training time. In fact, a new classifier is built for the target domain wherein the loss function is computed based on the target population, which lacks access to true class labels. As a result, the loss function is estimated using the source data by weighing the contribution of each source instance:

$$\hat{\theta} \equiv \arg\min_{\theta \in \Theta} \left[\frac{1}{n} \sum_{i=1}^{n} w_i \mathcal{L}(\hat{y}(x_i, \theta), y_i) \right], \tag{7}$$

where $\mathcal{L}$ represents the loss function between the predicted outcome $\hat{y}$ and the true label y. The weights $\{w_i\}_{i=1}^n$ are learned by means of an fairness-aware optimisation method, that aims to guarantee SP and low classification loss. The method constitutes an iterative learning framework where (i) for a given set of weights $w = \{w_i\}_{i=1}^n$ the optimal parameter values are learned (Equation (7)), (ii) then the most representative set of weights is estimated solving the fairness-aware optimization problem using that optimal parameter setting. This process is iteratively repeated until it converges to a stable solution. The main advantage of reweighing methods is that they are compatible with a diverse set of classification algorithms. However, it becomes problematic when the

probability distributions of the source and target domain have distinct support, a scenario that is widely common in real-world settings.

To overcome the weaknesses of instance weighing approaches, several authors have proposed to learn a new in-processing method for the deployment environment that jointly accounts for distribution shift and fairness by means of *adversarial learning* [44]: the adversary makes up the worst-case target distribution $\mathbb{Q}(x, a, y)$ that is coherent with the available target data, and the learner aims to find the parameter setting that optimises the predictive performance and fairness guarantees under such distribution. This can be seen as a minimax optimisation problem. Besides, note that, in a fairness-aware scenario, the worst case is defined not only by the predictive performance but also by fairness guarantees. Whenever fairness is achieved under the worst case, the model's fairness is guaranteed on the test data. A variety of shifted scenarios have been proposed inside this framework where, for each case, $\mathbb{Q}(x, a, y)$ is defined differently. For instance, Rezaei et al. [92] considered a general covariate shift scenario, and the worst-case distribution $\mathbb{Q}$ that approximates the target distribution is constrained to match a set of statistics of the source and target distributions. The constraints w.r.t. to the target data account for fairness requirements, and their characterisation varies for the different notions of fairness. Specifically, they characterize the constraints for the notions of SP, EOp, and EO. Du and Wu [30] adopted a similar approach but addressed the particular case when covariate shift is a consequence of sample selection bias, i.e., when the source instances are a sub-population of the target environment that have been labelled under a biased labelling policy, which is unknown. Therefore, in this scenario, the adversary suggests the sample selection probability function that leads to the worst-case scenario, while also aiming to achieve SP.

Moreover, An et al. [3] introduced the inclusion of an extra regularization term into the objective function of in-processing fair methods. These methods typically involve fairness and classification losses and the additional term serves the purpose of transferring fairness to the target domain. They introduced a self-training framework in which a model undergoes iterative training, and fairness is transferred to the target domain through consistency training [104] using data from both domains. The algorithmic fairness is evaluated in terms of EO and the variance of group accuracy.

(Scenario 4): Unsupervised adaptation without access sensitive information and $A_s = A_t$. This scenario is similar to the previously addressed, but the main difference between them is that, in this case, there is no access to the sensitive information in the target domain: that is, only X is known about the target data. Consequently, the evaluation of fairness criteria on the target environment is not trivial. Coston et al. [22] proposed an in-processing reweighing method equivalent to that of the previous section (Equation (7)), but with a different weight learning procedure, since the sensitive information is only available in the source domain. In fact, they aim to learn the weights that are as close as possible to the conventional covariate shift weights (defined only by the non-sensitive attributes, available in both domains) subject to SP constraints on the source data (where the sensitive information is available).

Up until now, the presented scenarios and corresponding methods have assumed that the shifted environment contains the same sensitive information as the source domain (i.e., $A_s = A_t$). However, there may arise situations where the method needs to be adapted to a new environment where the sensitive information differs (i.e., $A_s \neq A_t$). In the following section, we discuss approaches that address such adversities during deployment. In most cases, these scenarios are tackled using in-processing approaches that learn a new data representation invariant to the domain and sensitive information. Additionally, this representation should be informative, meaning that a classifier trained on top of this representation should make accurate predictions. As before, different scenarios will be identified based on the assumed availability of target information for adapting the classifier.

(Scenario 5) Supervised adaptation with available sensitive information and $A_s \neq A_t$.

In this scenario, labelled data with sensitive information is available from the target environment, where the sensitive attribute is different from that of the source. This information might seem enough to build a new model for the deployment environment from scratch. However, the works that address this particular scenario focus on the case where target data itself is scarce and, therefore, insufficient to build a completely new model. Thus, to overcome such adversity, they propose a framework to benefit from a dataset belonging to a distinct population but having abundant labelled data to build a suitable model, even if this auxiliary dataset has a different sensitive attribute. For instance, in image processing tasks, developing countries tend to have scarce images due to limitations in their resources and, consequently, models behave poorly [99]. These countries could benefit from images of wealthier countries with better imaging resources to build fair and accurate models, even if the populations differ notably and the sensitive information is different due to socioeconomic factors (e.g., in one country race might represent a critical issue, while in another country the discrimination is deemed w.r.t. gender). For that purpose, Schumann et al. [96] proposed to learn a new representation of the data $h(x, a, y)$ and build a classifier f that when trained on the new representation $h(x, a, y)$ of the data will achieve high predictive performance and fairness guarantees on the source and target domains, even when the sensitive attribute (and therefore the sub-population grouping) changes. To achieve this goal, they employed an adversarial learning framework, where the objective function incorporates three losses: (1) the classification loss incurred by training a classifier on the new representation $h(x, a, y)$, as well as losses that consider (2) the adversary's ability to predict the sensitive attribute of the source data and (3) the adversary's ability to predict the domain to which instances belong when using the new representation $h(x, a, y)$. Algorithmic fairness in this method is assessed through EO and EOp.

(Scenario 7) Unsupervised adaptation with available sensitive information and $A_s \neq A_t$.

This scenario is similar to the previous but less restrictive: label information is assumed to be unavailable from the target data, and is therefore more difficult to address. To find a data representation that is invariant to both the domain and the sensitive information in such scenario, Yoon et al. [119] proposed an approach based on the Optimal Transport framework. As a result, the new data representation aims to minimize the dissimilarities between the distributions of: (1) various sensitive groups in the source domain, (2) privileged populations across domains, and (3) unprivileged populations across domains. Additionally, a classifier trained on this representation is expected to make accurate and fair predictions in both domains, based on either SP or EOp. The distance between distributions is measured by the Wasserstein distance, which has many advantages over other existing divergence measures (such as, total variation distance, Kullback-Liebler divergence, and Jensen-Shannon divergence), since it takes into account the underlying geometry and assigns a finite divergence value even if the two distributions do not share support.

4.1.2 Distribution-based Fair Adaptive Approaches (A.2). Some approaches consider the scenario in which there is knowledge about how the target distribution is shifted from the source distribution, i.e., they assume certain knowledge about $\mathbb{P}_t(x, a, y)$. For instance, Giguere et al. [41] considered that the distribution shift is caused by *demographic shift*, and assumed that the demographic proportions (i.e., $\mathbb{P}_t(a)$) are known in the target environment. They contemplate two degrees of knowledge: first, the case where the demographic proportions are known explicitly, and second, when such proportions are known to be bounded by some intervals. They built their model grounding on Seldonian algorithms [106]. They divided the training data into two blocks: the first block is used to train a fair algorithm; however, the data from second block is used to compute a high confidence upper bound of the fairness guarantees of the fair algorithm in the deployment

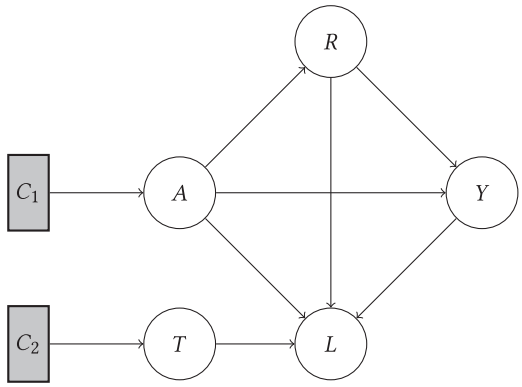


Fig. 2. Joint causal graph that describes the influence of the distribution shift on the features. C_1 and C_2 are the context variables that describe the effect of distribution shift, i.e., the distributions of which variables are going to change because of the shift. R, T, and L are non-sensitive features, A is the sensitive feature, and Y the class label.

environment. This upper bound incorporates the demographic information that is known about the target environment (exact proportions or bounds). If the fair algorithm is expected to behave fairly with high confidence on the target environment, then their algorithm returns the initial fair model; otherwise, no model is provided. Being a versatile training framework, it can be employed in conjunction with any procedure for training a fair classifier and applied to any fairness criteria.

4.1.3 Fair Causal Approaches (A.3). In general, causal approaches consider that there is no access to the target domain data, but presume the whole data generating process is described by means of a causal graph and the latter is known [101, 102]. Particularly, they use the *joint causal inference* framework [76], a modelling approach that unifies the data generating process for all the distinct domains as a single causal graph that represents the underlying causal model. Figure 2 shows one such model. This framework introduces *context variables* C_i, which describe how the different domains differ. In this way, they provide a complete overview about the plausible distribution shifts through the edges connecting the context variable(s) and the features. This information is highly relevant for the construction of classifiers that do not deteriorate under arbitrary shifts. Particularly, the key step of these causal approaches is to find a feature subset $S \subseteq \{X, A\}$ that satisfies the following assumptions:

- **Assumption 1: Invariance of the classification error.** S is a separating set, i.e., the conditional distribution of the class label Y given the features S is invariant across the different domains: $Y \perp\!\!\!\perp C \mid S$
- **Assumption 2: Invariance of the fairness constraints.** Initially, Singh et al. [101] introduced one variation of this assumption specifically for the EOp notion. However, in their subsequent work [102], they extended this assumption to encompass other fairness notions such as SP, EO, and predictive equality.

These assumptions ensure that both the error and fairness constraints are identifiable in the target domain. When S is identified, a classifier is built on top of it so that it does not deteriorate when deployed in any environment accounted in the causal graph:

$$\hat{\theta} \equiv \arg\min_{\theta \in \Theta} \mathbb{E}_{(s,y)\sim P_t}[\mathcal{L}(\theta, S, Y) \mid \boldsymbol{g}(\theta, S, Y) \leq \epsilon], \tag{8}$$

where $\mathcal{L} : \Theta \times S \times Y \longrightarrow \mathbb{R}$ is the loss function and $\boldsymbol{g}(\theta, S, Y)$ is the function denoting the fairness constraints. Thus, this optimisation problem aims to find the parameter setting that minimises the classification error in the target domain while satisfying statistical fairness notions.

The approaches based on causal frameworks suffer from several drawbacks: on the one hand, they require the knowledge of the causal graph that describes the data generating process; on the other hand, in general, the direct effects of the context variables on the rest of the features, as well as those between the features itself, are not trivial to define [95]. Further, these methods rely on strong shift assumptions, which limit their deployment in real-world applications, where those premises are hardly hold. Consequently, in most cases, adopting this framework leads to solutions with no guarantees or trivial predictors.

4.2 Fair Robust Approaches

The above presented methods overcome the drawbacks caused by distribution shift for particular deployment environments. However, those solutions can be highly sensitive to small perturbations, and an optimal solution can turn into a highly sub-optimal or even infeasible solution. Thus, it is essential to design fair robust approaches providing fair models that perform accurately on any possible unknown target environment and whose solutions remain nearly optimal under small perturbations of the data. Most of the fair robust approaches model the uncertainty over the deployment environment by means of an *ambiguity* or *uncertainty set* composed of perturbations of the empirical source distribution [30, 31, 71, 91, 105]. In this framework, known as ***distributionally robust optimisation*** **(DRO)**, models are made robust by finding the parameter setting that minimises the worst-case loss among all the probability distributions in the ambiguity set. Nonetheless, there are several approaches that follow distinct robustifying procedures [10, 111].

4.2.1 Distributionally Robust Optimisation (DRO). These approaches assume that the possible deployment environments belong to an *ambiguity* or *uncertainty set* $\mathcal{U}$ and propose to train the models by means of an optimisation problem defined over the parameter setting that minimises the worst-case loss among all the probability distributions in $\mathcal{U}$. This is usually formulated as a minimax problem in which the loss in minimised with respect to the worst-case realisation of the perturbations of the ambiguity set:

$$\min_{\theta \in \Theta} \max_{\mathbb{Q} \subset \mathcal{U}} \mathbb{E}_{(x,a,y)\sim\mathbb{Q}}[\mathcal{L}(\theta, X, A, Y) \mid \sup_{\mathbb{Q} \subset \mathcal{U}} \boldsymbol{g}(\theta, X, A, Y) \leq 0], \tag{9}$$

where $\mathcal{L} : \Theta \times X \times A \times Y \longrightarrow \mathbb{R}$ is a suitable loss function and $\boldsymbol{g}(\theta, X, A, Y)$ are the functions defining the constraints (when applicable). Figure 3 shows a simplified scheme of DRO approaches. The following is a classification of the main DRO approaches according to the ambiguity set they consider to model distributional uncertainty, which are summed up in Table 3. It is important to note that all of them assume that the sensitive attribute will remain unchanged for the possible unseen deployment environments.

Wasserstein ball. Taskesen et al. [105] claimed that the distribution describing the true data generating process is unknown and only indirectly observable through the training samples. They propose an ambiguity set that refers to the intersection between all the distributions within a Wasserstein ball centred at the empirical distribution and the set of all distributions under which the marginal of (A, Y) matches the empirical marginal. The latter eliminates unrealistic data distributions from the ambiguity set. By taking the worst-case approach they propose to robustify a fair in-processing method (where fairness concerns are added as a penalisation in the objective function) against all the distributions in the set that, with high probability, will contain the true distribution. However, Du and Wu [30] proposed a slightly different approach. They considered the special case where the distribution shift is caused by sample selection bias. Since the true sampling

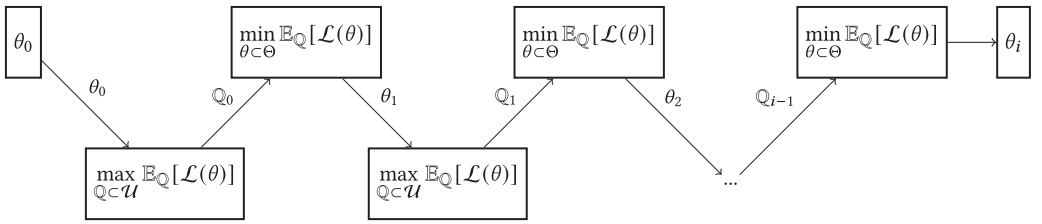


Fig. 3. Scheme of the general workflow of fair DRO approaches that take into account the uncertainty in the underlying probability distribution of the data. They aim to find a solution that remains effective across a range of possible distributions represented by an *ambiguity* or *uncertainty set* $\mathcal{U}$. The objective is to minimize the worst-case loss among all the probability distributions in $\mathcal{U}$ by means of an iterative procedure that involves alternately minimizing the worst-case loss over possible distributions in the uncertainty set and maximizing the worst-case loss over all choices of parameter settings. This iterative approach aims to find a robust solution that performs well across a range of potential data distributions, enhancing the solution's robustness against model misspecification and unanticipated data variations. The initialization (i.e., setting θ_0) can be accomplished using different procedures, including random initialization, pre-training, and problem-specific techniques.

Table 3. Characteristics of the Most Representative Fair DRO Approaches in Terms of the Defined Ambiguity Set, the Loss Function, How Fairness Is Accounted For and the Fairness Notions That Can Be Considered

Method	Ambiguity set	Loss	Fairness	Fairness criteria
[105]	Wasserstein ball centered at the empirical distribution	Log-loss	Weighted penalty	log-probabilistic + { EO, EOp, .. }
[30]	Wasserstein ball centered at uniform cluster selection ratio	Selection probability weighted loss	Constraint	Decision boundary fairness [122]
[91]	Statistics matching with the training data	Log-loss	Constraint	EO, EOp, DP, etc.
[71]	Weighted perturbations of the training dataset	Weighted empirical risk	Weighted penalty	EO, EOp, DP, etc.
[31]	Weighted perturbations of the training dataset	Agnostic	Constraint	Decision boundary fairness [122]

policy is unknown, they robustify the in-processing model w.r.t. to the worst possible sampling policy that could be encountered. That is, in this case, the uncertainty set contains the possible sampling policies. In particular, they cluster the data and assume that the instances belonging to the same cluster have the same sampling probability. Besides, the uncertainty set is a Wasserstein ball around the uniform selection ratio (that is, the case where all the clusters have the same selection probability). Moreover, different from the previous approach, fairness concerns are added as constraints that need to be satisfied in both the minimisation and the maximisation problems.

Weighted perturbations. Mandal et al. [71] developed a double mini-max iterative algorithm where the ambiguity set is defined by weighted perturbations of the empirical training distribution. In each iteration, the weighted perturbation that maximises the loss is chosen first. Then, the algorithm aims to learn the parameter setting that minimises the weighted empirical risk of the given perturbation and is fair with respect to the worst possible fairness violation that can be found in the ambiguity set (and, consequently, fair with respect to any perturbation in the ambiguity set). With this, they obtain a classifier that is fair and robust to any weighted perturbation of the training set. Du et al. [31] also proposed an approach that is robust to instance-level weighted perturbations of the original dataset. They assume these weighted perturbations are linear combinations of some

basis functions, e.g., Gaussian kernels. Fairness is enforced by means of additional constraints in both steps of the mini-max optimisation problem: the inner maximisation problem aims to find the perturbation that maximises the loss subject to fairness constraints, while the outer minimisation aims to find the parameter setting that minimises the worst-case loss subject to those constraints.

Statistics matching. Rezaei et al. [91] propose to modify an in-processing method, where fairness guarantees are added as constraints, with the objective of achieving robust performance under covariate shift. In this approach, the ambiguity set is composed by all the distributions that match the statistics of the training data, characterised by a vector valued feature function. This robust model is suitable for a wide range of fairness notions, such as DP, EO, and EOp.

4.2.2 Deployment Environment Agnostic Procedures. Although the DRO framework has been most popularly considered when building fair and robust models, there are also few works that do not make any assumptions about the unknown deployment environment. For instance, Wang et al. [111] aimed to build a fair model for which the curvature of the loss function is equivalent for all the sensitive groups. With that, the model will have similar robustness across the different sensitive groups, i.e., the generalisation capability of the model will be equivalent for all the data that is not supported in the source distribution, independent of the sensitive attribute. The base fair model is trained using a conventional fair adversarial training method, which consists of a classification and fairness loss. Additionally, they incorporate an extra component in the loss function to consider the similarity of the loss curvature across different sensitive groups, measured using **Maximum Mean Discrepancy (MMD)** [45]. Specifically, they report algorithmic fairness in terms of EO. However, Biswas and Mukherjee [10] proposed a robust model to address label shift (i.e., shift in the prevalence). In particular, they proposed to build decoupled classifiers for each sensitive group. Each of those classifiers has two constituent blocks: (1) a quantifier to estimate the prevalence of a given dataset (with datapoints that belong to sensitive group a) and (2) an ensemble classifier. Every base classifier of the ensemble is trained with a randomly selected subset of the original dataset (of instances belonging to a), where each of those subsets will have a different prevalence. Consequently, each of the base classifiers of the ensemble is specialised in providing accurate prediction in datasets with a given prevalence. Therefore, when the classifier is deployed in a unknown deployment environment, first the target data is divided into different sensitive groups. Then, for each sensitive group they estimate the prevalence using the quantifier, and finally, they use the base classifier of the ensemble of a that has been trained with data having the most similar prevalence. Moreover, they introduce a novel fairness concept known as *prevalence difference*, which captures the disparity between the actual and predicted prevalences for each sensitive group in the altered environment.

5 DATASETS AND BENCHMARKING

In the experimental section, many methods use popular datasets belonging to the literature of algorithmic fairness and create synthetic shifts to evaluate the proposed model. However, there are many datasets suitable for benchmarking purposes that naturally contain distribution shifts that arise in the real world as well as sensitive information. In what follows, we briefly describe the most popular datasets used in the literature of this research area.

American Community Survey (ACS) data [29]. This dataset is an extended version of the well-known UCI Adult dataset, which originates from publicly accessible data provided by the U.S. Census. It serves as an exceptionally suitable collection for evaluating the efficacy of methods that simultaneously address algorithmic fairness and distribution shift, making it a valuable resource for performance assessments in this research domain. It covers different prediction tasks related to income, employment, health, transportation and housing. For each of the tasks they provide data belonging to all fifty U.S. states and Puerto Rico, and five years of data collection between 2014 and

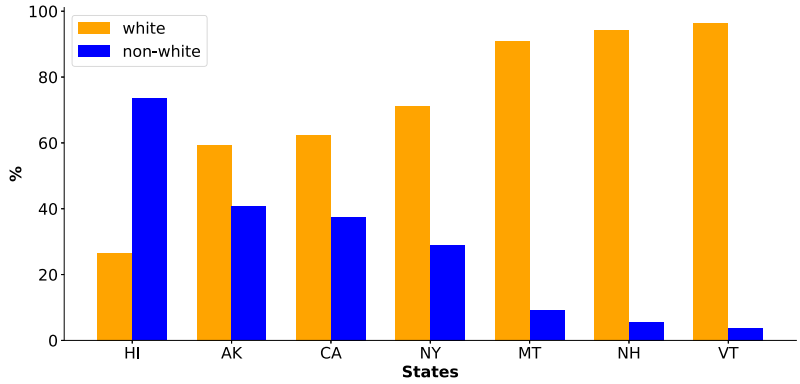


Fig. 4. Subpopulation shift in the ACSEmployment dataset. This phenomenon highlights substantial variations in the proportion of individuals from diverse racial backgrounds across different states. The figure displays 7 of 50 states, including 3 with the highest proportion of non-white population (HI, AK, CA), 3 with the lowest proportion of non-white population (MT, NH, VT), and 1 state with an intermediate situation (NY). The dataset underscores regional differences in demographic composition, impacting employment trends, and offering crucial insights for equitable policy-making. For an in-depth analysis of the race distribution in two specific states (HI and VT) with the highest and lowest proportion of non-white population, respectively, please see Figure 5.

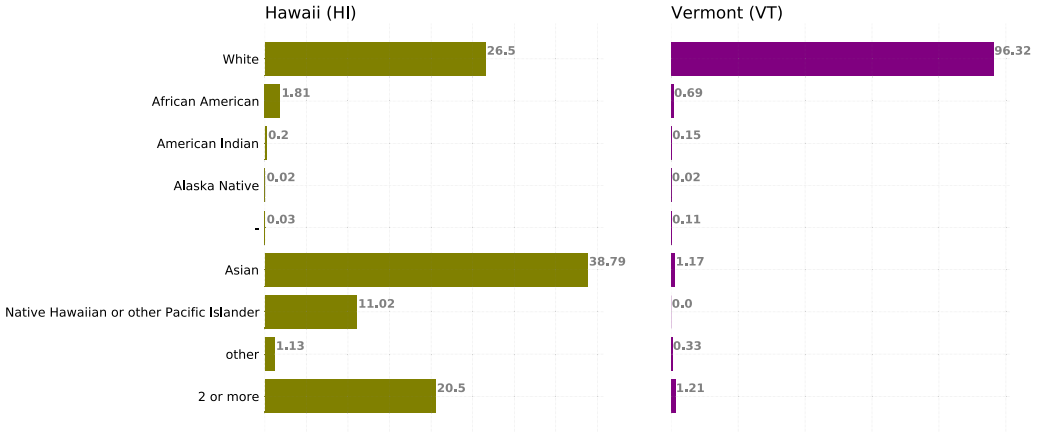


Fig. 5. Examining subpopulation shift in the ACSEmployment dataset. A visual analysis of the racial demographic composition in Hawaii (HI) and Vermont (VT), representing states with the highest and lowest proportion of non-white population, respectively. The figure accentuates significant variations in the proportion of individuals from diverse racial backgrounds across these states.

2018, both inclusive. That is, for each prediction task there are 255 distinct datasets available to account for real geographical and temporal variations (see Figures 4–7). Further, it is also possible to concatenate each state's data to create datasets that enclose all the U.S. Besides, all the the tasks have sensitive information enfolded in the features for age, race, and sex. This structure enclosing multiple tasks and shift is collected in the Python package `folktables` [28]. This set considers multiple prediction tasks, however, the main ones are listed below:

— **ACSIncome:** this task aims to predict whether an individual's income exceeds the amount of 50,000$. The dataset comprises individuals who are aged 16 or above, have reported a

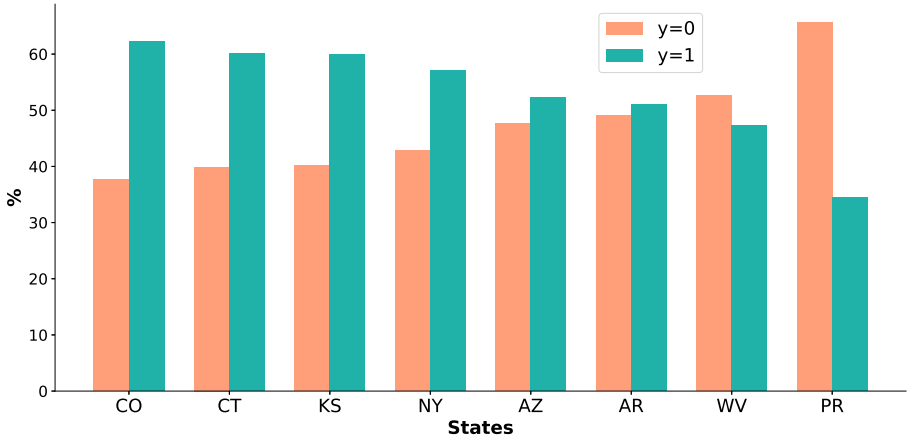


Fig. 6. Label distribution heterogeneity in the ACSEmployment dataset. This figure showcases the substantial variations in class label proportions across different states, shedding light on the diverse employment trends across geographical regions. Included are 8 of 50 states, each presenting distinct class label proportions, ranging from that with the highest $P(y = 1)$ to that with the lowest.

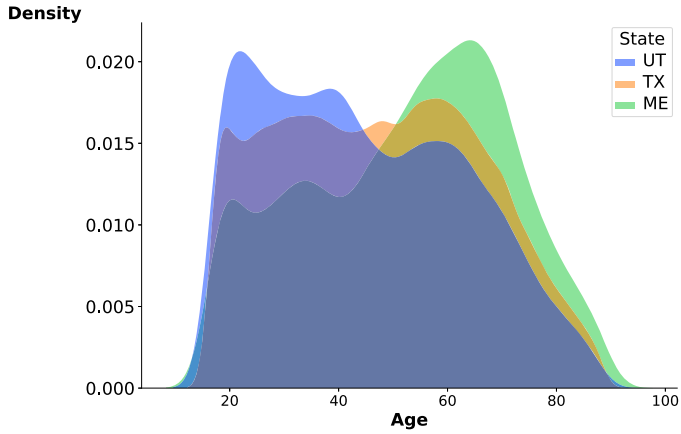


Fig. 7. Analyzing covariate shift in the ACSEmployment dataset. This visualization highlights the diverse age distribution of individuals across states, revealing the presence of covariate shift. Featured are the age distributions of 3 of 50 states: Utah (UT), Texas (TX), and Maine (ME). These states have the highest (ME), lowest (UT), and an intermediate (TX) expected value concerning age. The distinct patterns in age composition among these populations underscore the importance of considering demographic variations when studying employment trends and making data-driven decisions.

minimum of 1 hour per week of working activity in the preceding year, and possess an income of at least 100$.

— **ACSPublicCoverage:** the objective of this task is to correctly predict whether an low-income individual is covered by public health insurance. The samples are individuals under the age of 65 whose income is lower than 30,000$.

— **ACSMobility:** this prediction task focuses on whether an individual aged between 18 and 35 had the same residential address one year ago.

— **ACSEmployment:** aims to predict whether an individual aged between 16 and 90 is employed.
— **ACSTravelTime:** to goal of this task is to predict whether the commute to work is higher than 20 minutes for individuals that are older than 16.

WILDS [60]. WILDS is a collection of 10 datasets that describe a wide range of distribution shifts based on real-world applications, such as, shifts across hospitals for tumor identification. These datasets have three common characteristics: the shifts produce notable degradation on conventional ML classifiers that do not account for the shift, they reflect real-world settings and although they cannot be trivial, they must be solvable. Besides, WILDS contains a diverse set of tasks, data modalities, dataset sizes, and numbers of domains. This collection of datasets has been widely used in the field of fairness-agnostic distribution shift; however, several datasets contain sensitive information (e.g., CAMELYON17-WILDS, CIVILMODEL-WILDS, POVERTYMAP-WILDS, AMAZON-WILDS) thus are suitable for the analysis of classifier performance on a bias-aware distribution shift scenario.

MIMIC-III [54, 114]. Medical Information Mart for Intensive Care III (MIMIC-III) is a open access critical care dataset. It contains electronic health records (EHRs) of 53,423 patients (aged 16 or above) and 7,870 neonates admitted to intensive care units (ICUs) at the Beth Israel Deaconess Medical Center hospital between the years 2001 and 2012 (both inclusive). In particular, it collects data from 5 different ICUs: Medical ICU (MICU), Surgical ICU (SICU), Trauma Surgical ICU (TSICU), Cardiac Surgery Recovery Unit (CSRU) and Coronary Care Unit (CCU). The variables that characterize each patient include static features (e.g., age, admission type), time-series variables related to vital signs (e.g., heart rate, blood pressure) and laboratory test results (e.g., white blood cell counts). Further, it contains sensitive information such as age, race and gender. This dataset can be used for different tasks [86], including mortality prediction or estimation of the length of the stay at the ICU. Each of the ICUs can be viewed as a different domain or environment, since the population of patients varies from one ICU to others. It is worth highlighting that a recent release, MIMIC-IV, offers an updated version of the MIMIC-III dataset. This latest iteration provides researchers with an enhanced and more up-to-date collection of data, presenting new opportunities for analysis and investigation in the field.

6 RELATED RESEARCH AREAS

In this section, we describe other research areas that share some similarities with this field. Some of these areas enclose approaches that aim to preserve fairness guarantees in other non-standard albeit realistic scenarios. Besides, other fields are even more general and go beyond the conventional fairness-aware framework. Apart from those described below, we have seen that there are emerging connections, as yet little explored, but also worth mentioning such as robustness disparity in Reference [80].

6.1 Fair Inductive Transfer Learning

Learning fair classifiers under distribution shift is a particular framework of *fair transfer learning*: indeed, it constitutes a *transductive* transfer learning scenario where although the data generating process changes across domains, the tasks are the same in both environments. Nonetheless, there are other transfer learning scenarios, such as, *inductive* transfer learning, in which the tasks are different in the source and target domains. The main purpose of these frameworks is to benefit from combining previous knowledge about a collection of fairness-related tasks to obtain better and faster solutions, and requires access to labelled data. Using standard bias-agnostic transfer learning algorithms has proven to improve predictive accuracy on the target task at the cost of lowering the fairness guarantees of the predictions, a phenomenon called *discriminatory transfer*

[64]. Nevertheless, preliminary work in Reference [124] suggests that fairness can indeed be transferable when is explicitly accounted for, and has been demonstrated to do so in recent work [82, 103, 130, 131, 133, 134].

A popular fair inductive transfer learning setting is **fair meta-learning** [103, 131, 133, 134], in which the main objective is to build a model that can be easily adapted to a new task with minimal data and iterations. In particular, most of the fair meta-learning approaches focus on the few-shot learning problem: these methods aim to learn a general fair and accurate model (meta-model) that is valid as an initial model for multiple tasks, enabling the classifiers for new tasks to be learned with minimal data and a small number of gradient update steps. Furthermore, Zhao et al. [132] extended it to an online learning setting.

Multi-task learning is another inductive transfer learning framework. The key idea is to learn a set of relates tasks jointly to improve the predictive accuracy and fairness guarantees of the tasks as well as the generalisation guarantees. Although seemingly equivalent, the main purpose of the meta-learning and multi-task frameworks are indeed different: meta-learning aims to build a model that can be easily adapted to a new task with minimal data and iterations; however, multi-task learning aims to learn a shared representation to improve the predictive accuracy and fairness. Oneto et al. [82] proposed a multi-task learning framework to learn a representation that is fair as well as transferable. Further, the representation learned under such setting is proven to generalise well to new unseen tasks with regards to predictive performance and fairness. Moreover, Zhao and Chen [130] presented a fairness learning approach for multi-task regression models.

The fields of fair meta-learning and multi-task, as well as the fair methods that address distribution shift are concerned with the transfer of fairness guarantees. In the case of the former, the transfer is made across different but related tasks (where distribution shift can also happen); while for the latter, fairness guarantees are transferred across domains.

6.2 Federated Learning

Federated Learning (FL) [72] is a new ML framework to train models in a collaborative manner while protecting user privacy, data security, and government regulatory requirements. Each of the data owners trains a local model on local training data, and each of the owners shares the parameters of its model to generate a global model. For example, automated vehicles process large amounts of information locally using ML models, while working cooperatively with other vehicles and computing centres. In practice, the local training data of each owner may follow a different distribution, causing the global model to have poor performance on testing data. Therefore, the distribution shift between the different data collections cannot be ignored when building the global model to achieve high accuracy and fairness guarantees when deployed on local data or previously unseen environments. Besides, the amount of data that is generated and collected varies considerably for every client (or device). Therefore, fairness plays an important role in FL [31, 100, 136] not only related to data privacy or the discriminating behaviour of the model w.r.t. the sensitive information but also in how the collaboration between different agents is made.

Approaches grounded on federated learning need to address concerns related to distribution shift and fairness, albeit differently from those of the surveyed field: on the one hand, because model parameters are shared instead of the local data; and, on the other hand, because fairness concerns are not restricted to predictions, but encompass many other dimensions, such as, fairness in the collaboration when building the global model.

6.3 Fairness-aware Learning Under Corrupted Data

In many real-world settings the training and validation data are oftentimes unreliable and biased, e.g., when dealing with demographic data gathered through surveys or online polls. The presence

of errors, noise, or anomalies in the training data is denoted as data corruption, and can occur due to measurement errors, labelling mistakes, or data transmission issues, among other factors. Fair methods have shown to be vulnerable to data corruption [73] and, in the light of this issue, there is an emerging research line concerned about the development of fair classifiers that are robust to such adversity. Many works have been proposed to train fair models under noisy or adversarially perturbed sensitive attributes [5, 15, 16, 63, 74, 113], labels [36, 52, 81, 112], or both [53]. Nonetheless, other works address more general data corruption scenarios [61, 93]. The goal of these methods is to learn robust, accurate, and fair patterns despite the presence of data imperfections. Preserving fairness guarantees under corrupted data is similar to doing so under distribution shift, since, for both scenarios, the training data is not representative of the real population from the deployment environment. However, while distribution shift addresses the discrepancy between training and testing data distributions, data corruption deals with handling errors or anomalies present within the training data itself.

6.4 Spurious Correlations

A spurious correlation is the dependence of a model's predictions on aspects of the input data that should actually play a minor role, called nuisance factors. Those unwanted correlations are consequences of shortcut learning, which refers to the scenario in which algorithms rely on features that are easy to interpret and highly predictive of the class label. This association becomes problematic if the relation between the nuisance factor and the class label is unstable (i.e., varies for different deployment environments). In such case, whenever the model is deployed in an environment where the relation between the nuisance factors and the label is different (i.e., there exists distribution shift), it performs even worse than random guessing [43]. For example, in an animal classification task using natural images, the background (e.g., grass, snow, water) can be highly predictive of the animal. As a consequence, the model, which apparently predicted cows accurately, failed considerably when tested on pictures where cows appeared outside the typical grass landscape [8]. Several works have studied this difficulty and many approaches have been proposed to overcome it [40, 43, 70, 85, 108]. Even if it is a general framework, it could be applied to a fairness-aware scenario, where sensitive attributes would constitute the nuisance factors.

6.5 Long-term Fairness

In online learning settings, where the deployment environment is constantly evolving, it is also crucial to be aware of the long-term behaviours of deployed ML algorithms and their potential consequences [25, 47, 66, 78]. Several works have studied the long-term impact of conventional static fairness criteria on the well-being of the different sensitive groups [25, 66, 78]. In the light of the potential concerns that arise, others have proposed fairness-enhancing policies that ensure fairness/well-being in the long term [39, 47]. The fair online learning approaches that have been considered in the survey do not account for long-term fairness concerns, but rather focus on satisfying fairness guarantees on the current population. Nonetheless, as an extension to those works, it would be beneficial to combine both fields and create adaptive methods that follow a training procedure that accounts for long-time fairness guarantees.

7 DISCUSSION AND FUTURE RESEARCH CHALLENGES

ML research is facing increasingly complex problems. The task of obtaining reliable predictive algorithms, making use of statistical models, has been compounded by the challenge of dealing with changing environments. An additional difficulty arises from the fact that these algorithms operate by learning from potentially biased data, in which subgroups inevitably exist, so that ethical considerations must be taken into account to ensure equitable treatment. Initially, despite their

different nature, both problems have been treated separately in the literature with similar methods providing satisfactory results. However, when trying to tackle both at the same time, the former no longer works and specific methods to address distribution shift under fairness constraints are needed. Indeed, the continuous evolution intrinsic to our nature and society has given rise to the necessity of developing fair ML systems that preserve fairness guarantees when the data generating processes differ between the source (train) and target (test) domains. The reason for such a change in the environment, generally called distribution shift, can be very diverse and thus it is important first to have a classification of the different shifts considered in the literature so far, which was presented in Section 2.3.

The approaches that preserve fairness guarantees under distribution shift are highly relevant for many real-world applications involving geographical and temporal data. For instance, healthcare applications often face distribution shift scenarios, and for such critical settings, it is essential to provide robust fairness guarantees [17, 18, 35, 95, 116, 123]. Moreover, adaptive approaches could also overcome the disparities derived from the existing gap in imaging resources between countries [99]. Finally, it is worth noticing that there are many applications where the data gets easily outdated, and the underlying data generating process varies for future periods of time [25, 128]. For example, financial markets are highly dynamic and subject to various factors such as economic conditions, geopolitical events, and investor sentiment. The data used for training ML models in this domain can quickly become outdated as new market data becomes available. Moreover, the underlying dynamics of the financial markets can change over time due to shifts in market trends, regulatory changes, or the emergence of new trading strategies. To avoid rebuilding the predictive model from scratch using the most recent labelled data, which is often expensive or impossible to recollect for such purposes, hard efforts can be avoided by adapting the existing model to the new environment.

This survey tried to contribute to the ML literature with the proposal of a novel taxonomy of the existing approaches that address robustness in performance while guaranteeing fairness under distribution shift. Depending on whether there is information available about the deployment environment, the methods have been classified into two main groups: adaptive and robust methods. Among the adaptive family, we have identified three subgroups of methods depending on the type of information that is available about the target environment: data-based, distribution-based, and causal approaches. Furthermore, we have proposed a reasonable and even more exhaustive classification of the existing data-based methods in the first subgroup, which has resulted in eight different scenarios depending on the assumptions about the class label and sensitive information of the target data. However, fair robust approaches are designed to preserve fairness guarantees when the deployment environment is completely unknown. The majority of these works follow a DRO-based approach and model the uncertainty over the deployment environment by means of an *ambiguity set* composed of perturbations of the empirical source distribution. In addition to these, we have also found others that do not follow this specific framework and make no assumptions regarding the environment.

In Section 5, the survey described different datasets and benchmarking alternatives considered in the works within the field of study. Additionally, in Section 6, the survey highlighted the connection to other related research fields. By establishing these connections and addressing the remaining open problems, it becomes evident that there are still significant areas that require further investigation. As a result, we identify potential future research directions:

Theoretical guarantees. Simultaneous to the proposal of methods to overcome model degradation in distribution shift scenarios, several works study the theoretical guarantees of the proposed solutions in terms of the generalisation bound for the error and fairness guarantees. Indeed, those works define an upper-bound of the generalisation loss of either the error [30, 119], the fairness

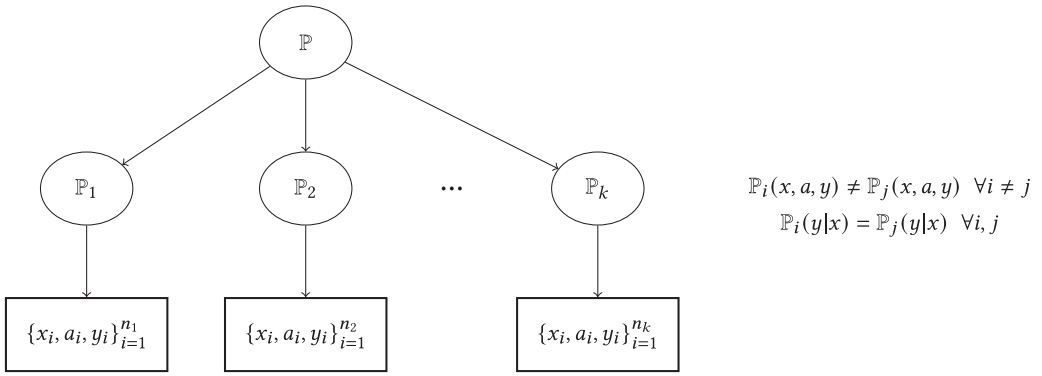


Fig. 8. Multi-source setting for fair domain generalization. The models designed for a fairness-aware domain generalization are trained on diverse datasets from multiple sources, representing various domains, to enhance its ability to generalize across unseen target domains. This strategy enables the model to achieve improved performance and adaptability when facing new and challenging data distributions.

guarantees [41, 96], or both [71] characterised by the training loss and/or unfairness, model complexity, and confidence. Besides, Chen et al. [19] generalise the theoretical guarantees of fairness transferability for any ML algorithm. However, most of the approaches on the field do not provide theoretical guarantees. In the light of this void, we encourage the research community to include theoretical results in future approaches. Furthermore, the fairness transferability results obtained until now are limited to particular shifts or models; thus, an extension of those results to more general scenarios would be an important future venue.

Harmful shift detection techniques. One of the most important future venues is the development of techniques to detect harmful distribution shifts. While adaptive methods assume knowledge of existing shifts between training and deployment environments, little attention has been given to practical methods for detecting the presence of such shifts. Not all shifts necessarily degrade the performance of the employed method, making it crucial to focus on developing specific tests that can identify instances where a classifier's performance or fairness guarantees may not generalize well in the deployment environment. These tests would be particularly valuable if they could be performed using a small sample set from the target environment.

Uncertainty-aware evaluation framework of fair models. It is well known that in real-world settings many assumptions made in the evaluation of the fair models do not hold. Therefore, a new evaluation framework is required for the safe deployment of the proposed methods in such settings. Assuming access to multiple independent datasets is non-realistic, and an exhaustive evaluation of a classifier in all the possible shifted deployment environments is infeasible. However, building on the idea of DRO approaches, it is possible to build a model-agnostic test where the classifier is tested under the worst-case target distribution (shifted from the source distribution) to verify whether the classifier is safe to be deployed in real-world settings without previous knowledge of the real environment it may encounter.

Multi-source approaches. The works that are part of the research area surveyed in this article consider a single source domain. However, in many real-world cases one might have access to labelled instances belonging to different underlying data distributions that share the functional form (see Figure 8). Thus, building models that account for a multi-source setting is of primary interest, either if they are adaptive or robust approaches.

Fair robust approaches that account for $A_s \neq A_t$. Up to the author's knowledge, all the robust methods assume that the sensitive attribute will be the same in any possible deployment

environment, which limits the applicability of the methods. With that, it is important to develop robust methods that account for the possibility of variation on the sensitive attribute in the likely deployment environments.

Multi-dimensional sensitive attribute and intersectionality. All the methods proposed in the field assume that each instance is defined by a single sensitive attribute. Besides, in most cases, the sensitive attribute is presumed to have two possible values. However, the real-world settings in which the models are deployed are far more complex: individuals contain many overlapping sensitive attributes, such as, race, gender, sexual orientation, class, and disability, and are therefore affected by a combination of systems of power and oppression [37]. Consequently, the fair methods that account for distribution shift should go beyond the binary sensitive attribute, and further consider multi-dimensional sensitive attributes.

APPENDIX

A SUMMARY TABLE OF ADAPTIVE APPROACHES

Table 4. Summary of the Main Characteristics of the Studied Approaches to Preserve Fairness Guarantees of Classifiers in Changing Environments

Scenario	Proc.	Method	Fairness criteria	Fairness-enhancing intervention	Shift type
A.1 - SC1	Pre-	[50]	SP	Reweighing or massaging instances of stream	Compound
	In-	[125]	Cummulative SP	Fairness-enhancing information gain splitting criteria for tree construction	Compound
		[126]	Cummulative SP	Fairness-enhancing (i) information gain splitting criteria for tree construction, (ii) sampling, and (iii) weighted voting	Compound
		[127]	Cummulative SP	Flexible fairness-enhancing information gain splitting criteria for tree construction	Compound
	Post-	[49]	Cummulative SP, EOp and predictive equality [109]	Decision threshold adjustment	Concept
A.1 - SC3	In-	[22]	SP	Classification loss with fairness-aware weighing function	Covariate
		[92]	SP, EOp and EO	Fairness penalty term in objective function	Covariate
		[30]	SP	Constraint	Compound
		[3]	EO and variance of group accuracy	Fairness-aware loss function	Demographic, compound and general
	Post-	[55]	EOp and EO	Adjust decision threshold according to weighted (target) TPR and FPR	Compound
A.1 - SC4	In-	[22]	SP	Classification loss with fairness-aware weighing function	Covariate

(Continued)

Table 4. Continued

Scenario	Proc.	Method	Fairness criteria	Fairness-enhancing intervention	Shift type
A.1 - SC5	In-	[96]	EOp and EO	Fairness-aware loss function	Concept
A.1 - SC7	In-	[119]	SP and EOp	Fairness-aware loss	Concept
A.2	In-	[41]	Any	Fairness-aware loss to choose candidates and fairness test to choose among candidates	Demographic
A.3	In-	[101] [102]	EOp SP, EOp, EO and predictive equality	Constraint Constraint	General General

REFERENCES

[1] Robert Adragna, Elliot Creager, David Madras, and Richard Zemel. 2020. Fairness and robustness in invariant learning: A case study in toxicity classification. In *Proceedings of the Neurips Workshop on Algorithmic Fairness through the Lens of Causality and Interpretability*.

[2] Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna Wallach. 2018. A reductions approach to fair classification. In *Proceedings of the International Conference on Machine Learning*. PMLR, Stockholm, Sweden, 60–69.

[3] Bang An, Zora Che, Mucong Ding, and Furong Huang. 2022. Transferring fairness under distribution shifts via fair consistency regularization. *Adv. Neural Info. Process. Syst.* 35 (2022), 32582–32597.

[4] Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner. 2016. Machine bias. In *Ethics of Data and Analytics*. Auerbach Publications, 254–264.

[5] Pranjal Awasthi, Matthäus Kleindessner, and Jamie Morgenstern. 2020. Equalized odds postprocessing under imperfect group information. In *Proceedings of the International Conference on Artificial Intelligence and Statistics*. PMLR, 1770–1780.

[6] Niels Bantilan. 2018. Themis-ml: A fairness-aware machine learning interface for end-to-end discrimination discovery and mitigation. *J. Technol. Hum. Serv.* 36, 1 (2018), 15–30.

[7] Yahav Bechavod and Katrina Ligett. 2017. Penalizing unfairness in binary classification. Retrieved from https://arXiv:1707.00044

[8] Sara Beery, Grant Van Horn, and Pietro Perona. 2018. Recognition in terra incognita. In *Proceedings of the European Conference on Computer Vision (ECCV'18)*. 456–473.

[9] Rachel K. E. Bellamy, Kuntal Dey, Michael Hind, Samuel C. Hoffman, Stephanie Houde, Kalapriya Kannan, Pranay Lohia, Jacquelyn Martino, Sameep Mehta, Aleksandra Mojsilović et al. 2019. AI fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias. *IBM J. Res. Dev.* 63, 4/5 (2019), 4–1.

[10] Arpita Biswas and Suvam Mukherjee. 2021. Ensuring fairness under prior probability shifts. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*. AAAI Press, 414–424.

[11] Tolga Bolukbasi, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam T. Kalai. 2016. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. *Adv. Neural Info. Process. Syst.* 29 (2016).

[12] Marc-Etienne Brunet, Colleen Alkalay-Houlihan, Ashton Anderson, and Richard Zemel. 2019. Understanding the origins of bias in word embeddings. In *Proceedings of the International Conference on Machine Learning*. PMLR, 803–811.

[13] Toon Calders and Sicco Verwer. 2010. Three naive Bayes approaches for discrimination-free classification. *Data Min. Knowl. Discov.* 21, 2 (2010), 277–292.

[14] Flavio P. Calmon, Dennis Wei, Bhanukiran Vinzamuri, Karthikeyan Natesan Ramamurthy, and Kush R. Varshney. 2017. Optimized pre-processing for discrimination prevention. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*. 3995–4004.

[15] L. Elisa Celis, Lingxiao Huang, Vijay Keswani, and Nisheeth K. Vishnoi. 2021. Fair classification with noisy protected attributes: A framework with provable guarantees. In *Proceedings of the International Conference on Machine Learning*. PMLR, 1349–1361.

[16] L. Elisa Celis, Anay Mehrotra, and Nisheeth Vishnoi. 2021. Fair classification with adversarial perturbations. *Adv. Neural Info. Process. Syst.* 34 (2021), 8158–8171.

[17] Irene Y. Chen, Shalmali Joshi, Marzyeh Ghassemi, and Rajesh Ranganath. 2021. Probabilistic machine learning for healthcare. *Annu. Rev. Biomed. Data Sci.* 4 (2021), 393–415.

[18] Richard J. Chen, Tiffany Y. Chen, Jana Lipkova, Judy J. Wang, Drew F. K. Williamson, Ming Y. Lu, Sharifa Sahai, and Faisal Mahmood. 2021. Algorithm fairness in ai for medicine and healthcare. Retrieved from https://arXiv:2110.00603

[19] Yatong Chen, Reilly Raab, Jialu Wang, and Yang Liu. 2022. Fairness transferability subject to bounded distribution shift. *Adv. Neural Info. Process. Syst.* 35 (2022), 11266–11278.

[20] Alexandra Chouldechova. 2017. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data* 5, 2 (2017), 153–163.

[21] Sam Corbett-Davies, Emma Pierson, Avi Feller, Sharad Goel, and Aziz Huq. 2017. Algorithmic decision making and the cost of fairness. In *Proceedings of the 23rd ACM Sigkdd International Conference on Knowledge Discovery and Data Mining*. ACM, New York, NY, 797–806.

[22] Amanda Coston, Karthikeyan Natesan Ramamurthy, Dennis Wei, Kush R. Varshney, Skyler Speakman, Zairah Mustahsan, and Supriyo Chakraborty. 2019. Fair transfer learning with missing protected attributes. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 91–98.

[23] Elliot Creager, Jörn-Henrik Jacobsen, and Richard Zemel. 2020. Exchanging lessons between algorithmic fairness and domain generalization. arXiv preprint arXiv:2010.07249. https://arxiv.org/abs/2010.07249

[24] Jessica Dai and Sarah M. Brown. 2020. Label bias, label shift: Fair machine learning with unreliable labels. In *Proceedings of the NeurIPS Workshop on Consequential Decision Making in Dynamic Environments*, Vol. 12.

[25] Alexander D'Amour, Hansa Srinivasan, James Atwood, Pallavi Baljekar, David Sculley, and Yoni Halpern. 2020. Fairness is not static: Deeper understanding of long-term fairness via simulation studies. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. ACM, New York, NY, 525–534.

[26] Amit Datta, Michael Carl Tschantz, and Anupam Datta. 2015. Automated experiments on ad privacy settings. *Proc. Priv. Enhanc. Technol.* 2015, 1 (2015), 92–112.

[27] Emily Diana, Wesley Gill, Michael Kearns, Krishnaram Kenthapadi, and Aaron Roth. 2021. Minimax group fairness: Algorithms and experiments. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 66–76.

[28] Frances Ding, Moritz Hardt, John Miller, and Ludwig Schmidt. 2021. *folktables*. https://github.com/zykls/folktables

[29] Frances Ding, Moritz Hardt, John Miller, and Ludwig Schmidt. 2021. Retiring adult: New datasets for fair machine learning. *Adv. Neural Info. Process. Syst.* 34 (2021).

[30] Wei Du and Xintao Wu. 2021. Fair and robust classification under sample selection bias. In *Proceedings of the 30th ACM International Conference on Information and Knowledge Management*. 2999–3003.

[31] Wei Du, Depeng Xu, Xintao Wu, and Hanghang Tong. 2021. Fairness-aware agnostic federated learning. In *Proceedings of the SIAM International Conference on Data Mining (SDM'21)*. SIAM, 181–189.

[32] Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. 2012. Fairness through awareness. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*. 214–226.

[33] Cynthia Dwork, Nicole Immorlica, Adam Tauman Kalai, and Max Leiserson. 2018. Decoupled classifiers for group-fair and efficient machine learning. In *Proceedings of the Conference on Fairness, Accountability and Transparency*. PMLR, 119–133.

[34] Michael Feldman, Sorelle A. Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. 2015. Certifying and removing disparate impact. In *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 259–268.

[35] Samuel G. Finlayson, Adarsh Subbaswamy, Karandeep Singh, John Bowers, Annabel Kupke, Jonathan Zittrain, Isaac S. Kohane, and Suchi Saria. 2021. The clinician and dataset shift in artificial intelligence. *New Engl. J. Med.* 385, 3 (2021), 283.

[36] Riccardo Fogliato, Alexandra Chouldechova, and Max G'sell. 2020. Fairness evaluation in presence of biased noisy labels. In *Proceedings of the International Conference on Artificial Intelligence and Statistics*. PMLR, 2325–2336.

[37] James R. Foulds, Rashidul Islam, Kamrun Naher Keya, and Shimei Pan. 2020. Bayesian modeling of intersectional fairness: The variance of bias. In *Proceedings of the SIAM International Conference on Data Mining*. SIAM, 424–432.

[38] Andreas Fuster, Paul Goldsmith-Pinkham, Tarun Ramadorai, and Ansgar Walther. 2022. Predictably unequal? The effects of machine learning on credit markets. *J. Finance* 77, 1 (2022), 5–47.

[39] Yingqiang Ge, Shuchang Liu, Ruoyuan Gao, Yikun Xian, Yunqi Li, Xiangyu Zhao, Changhua Pei, Fei Sun, Junfeng Ge, Wenwu Ou et al. 2021. Towards long-term fairness in recommendation. In *Proceedings of the 14th ACM International Conference on Web Search and Data Mining*. 445–453.

[40] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. Shortcut learning in deep neural networks. *Nature Mach. Intell.* 2, 11 (2020), 665–673.
[41] Stephen Giguere, Blossom Metevier, Bruno Castro da Silva, Yuriy Brun, Philip S. Thomas, and Scott Niekum. 2021. Fairness guarantees under demographic shift. In *Proceedings of the International Conference on Learning Representations*.
[42] Gabriel Goh, Andrew Cotter, Maya Gupta, and Michael P. Friedlander. 2016. Satisfying real-world goals with dataset constraints. In *Advances in Neural Information Processing Systems*. 2415–2423.
[43] Mark Goldstein, Jörn-Henrik Jacobsen, Olina Chau, Adriel Saporta, Aahlad Manas Puli, Rajesh Ranganath, and Andrew Miller. 2022. Learning invariant representations with missing data. In *Proceedings of the Conference on Causal Learning and Reasoning*. PMLR, 290–301.
[44] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative adversarial nets. *Adv. Neural Info. Process. Syst.* 27 (2014).
[45] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. 2012. A kernel two-sample test. *J. Mach. Learn. Res.* 13, 1 (2012), 723–773.
[46] Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of opportunity in supervised learning. *Adv. Neural Info. Process. Syst.* 29 (2016), 3315–3323.
[47] Lily Hu and Yiling Chen. 2018. A short-term intervention for long-term fairness in the labor market. In *Proceedings of the World Wide Web Conference*. 1389–1398.
[48] Ben Hutchinson and Margaret Mitchell. 2019. 50 years of test (un) fairness: Lessons for machine learning. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. 49–58.
[49] Vasileios Iosifidis and Eirini Ntoutsi. 2020. FABBOO—Online fairness-aware learning under class imbalance. In *Proceedings of the International Conference on Discovery Science*. Springer, 159–174.
[50] Vasileios Iosifidis, Thi Ngoc Han Tran, and Eirini Ntoutsi. 2019. Fairness-enhancing interventions in stream classification. In *Proceedings of the International Conference on Database and Expert Systems Applications*. Springer, 261–276.
[51] Sophie Jentzsch, Patrick Schramowski, Constantin Rothkopf, and Kristian Kersting. 2019. Semantics derived automatically from language corpora contain human-like moral choices. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. 37–44.
[52] Heinrich Jiang and Ofir Nachum. 2020. Identifying and correcting label bias in machine learning. In *Proceedings of the International Conference on Artificial Intelligence and Statistics*. PMLR, 702–712.
[53] Changhun Jo, Jy-yong Sohn, and Kangwook Lee. 2022. Breaking fair binary classification with optimal flipping attacks. In *Proceedings of the IEEE International Symposium on Information Theory (ISIT'22)*. IEEE, 1453–1458.
[54] Alistair E. W. Johnson, Tom J. Pollard, Lu Shen, Li-wei H. Lehman, Mengling Feng, Mohammad Ghassemi, Benjamin Moody, Peter Szolovits, Leo Anthony Celi, and Roger G. Mark. 2016. MIMIC-III, a freely accessible critical care database. *Sci. Data* 3, 1 (2016), 1–9.
[55] Nathan Kallus and Angela Zhou. 2018. Residual unfairness in fair machine learning from prejudiced data. In *Proceedings of the International Conference on Machine Learning*. PMLR, 2439–2448.
[56] Faisal Kamiran and Toon Calders. 2012. Data preprocessing techniques for classification without discrimination. *Knowl. Info. Syst.* 33, 1 (2012), 1–33.
[57] Toshihiro Kamishima, Shotaro Akaho, Hideki Asoh, and Jun Sakuma. 2012. Fairness-aware classifier with prejudice remover regularizer. In *Proceedings of the Joint European Conference on Machine Learning and Knowledge Discovery in Databases*. Springer, 35–50.
[58] Thomas Kehrenberg, Myles Bartlett, Oliver Thomas, and Novi Quadrianto. 2020. Null-sampling for interpretable and fair representations. In *Proceedings of the 16th European Conference on Computer Vision (ECCV'20) (Lecture Notes in Computer Science, Vol. 12371)*, Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm (Eds.). Springer, 565–580.
[59] Amir E. Khandani, Adlar J. Kim, and Andrew W. Lo. 2010. Consumer credit-risk models via machine-learning algorithms. *J. Bank. Finance* 34, 11 (2010), 2767–2787.
[60] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanas Phillips, Irena Gao et al. 2021. Wilds: A benchmark of in-the-wild distribution shifts. In *Proceedings of the International Conference on Machine Learning*. PMLR, 5637–5664.
[61] Nikola H. Konstantinov and Christoph Lampert. 2022. Fairness-aware PAC learning from corrupted data. *J. Mach. Learn. Res.* 23 (2022), 1–60.
[62] Nikita Kozodoi and Tibor V. Varga. 2021. *Fairness: Algorithmic Fairness Metrics*. Retrieved from https://CRAN.R-project.org/package=fairness R package version 1.2.1.
[63] Alex Lamy, Ziyuan Zhong, Aditya K. Menon, and Nakul Verma. 2019. Noise-tolerant fair classification. *Adv. Neural Info. Process. Syst.* 32 (2019).

[64] Chao Lan and Jun Huan. 2017. Discriminatory transfer. In *Proceedings of the Workshop on Fairness, Accountability and Transparency in Machine Learning (FAT/ML'17)*.
[65] Zachary Lipton, Julian McAuley, and Alexandra Chouldechova. 2018. Does mitigating ML's impact disparity require treatment disparity? *Adv. Neural Info. Process. Syst.* 31 (2018).
[66] Lydia T. Liu, Sarah Dean, Esther Rolf, Max Simchowitz, and Moritz Hardt. 2018. Delayed impact of fair machine learning. In *Proceedings of the International Conference on Machine Learning*. PMLR, 3150–3158.
[67] Christos Louizos, Kevin Swersky, Yujia Li, Max Welling, and Richard Zemel. 2016. The variational fair autoencoder. In *Proceedings of the International Conference on Learning Representations*.
[68] David Madras, Elliot Creager, Toniann Pitassi, and Richard Zemel. 2018. Learning adversarially fair and transferable representations. In *Proceedings of the 35th International Conference on Machine Learning*, Vol. 80. PMLR, Stockholm, Sweden, 3384–3393.
[69] Subha Maity, Debarghya Mukherjee, Mikhail Yurochkin, and Yuekai Sun. 2021. Does enforcing fairness mitigate biases caused by subpopulation shift? *Adv. Neural Info. Process. Syst.* 34 (2021), 25773–25784.
[70] Maggie Makar, Ben Packer, Dan Moldovan, Davis Blalock, Yoni Halpern, and Alexander D'Amour. 2022. Causally motivated shortcut removal using auxiliary labels. In *Proceedings of the International Conference on Artificial Intelligence and Statistics*. PMLR, 739–766.
[71] Debmalya Mandal, Samuel Deng, Suman Jana, Jeannette Wing, and Daniel J. Hsu. 2020. Ensuring fairness beyond the training data. *Adv. Neural Info. Process. Syst.* 33 (2020), 18445–18456.
[72] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Proceedings of the International Conference on Artificial Intelligence and Statistics*. PMLR, 1273–1282.
[73] Ninareh Mehrabi, Muhammad Naveed, Fred Morstatter, and Aram Galstyan. 2021. Exacerbating algorithmic bias through fairness attacks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 8930–8938.
[74] Anay Mehrotra and L. Elisa Celis. 2021. Mitigating bias in set selection with noisy protected attributes. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency*. ACM, New York, NY, 237–248.
[75] Aditya Krishna Menon and Robert C. Williamson. 2018. The cost of fairness in binary classification. In *Proceedings of the Conference on Fairness, Accountability and Transparency*. PMLR, 107–118.
[76] Joris M. Mooij, Sara Magliacane, and Tom Claassen. 2020. Joint causal inference from multiple contexts. *J. Mach. Learn. Res.* 21, 99 (2020), 1–108.
[77] Jose G. Moreno-Torres, Troy Raeder, Rocío Alaiz-Rodríguez, Nitesh V. Chawla, and Francisco Herrera. 2012. A unifying view on dataset shift in classification. *Pattern Recogn.* 45, 1 (2012), 521–530.
[78] Hussein Mouzannar, Mesrob I. Ohannessian, and Nathan Srebro. 2019. From fair decision making to social equality. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. 359–368.
[79] Debarghya Mukherjee, Felix Petersen, Mikhail Yurochkin, and Yuekai Sun. 2022. Domain adaptation meets individual fairness. And they get along. *Adv. Neural Info. Process. Syst.* 35 (2022), 28902–28913.
[80] Vedant Nanda, Samuel Dooley, Sahil Singla, Soheil Feizi, and John P. Dickerson. 2021. Fairness through robustness: Investigating robustness disparity in deep learning. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency*. 466–477.
[81] Amanda Olmin and Fredrik Lindsten. 2022. Robustness and reliability when training with noisy labels. In *Proceedings of the International Conference on Artificial Intelligence and Statistics*. PMLR, 922–942.
[82] Luca Oneto, Michele Donini, Massimiliano Pontil, and Andreas Maurer. 2020. Learning fair and transferable representations with theoretical guarantees. In *Proceedings of the IEEE 7th International Conference on Data Science and Advanced Analytics (DSAA'20)*. IEEE, 30–39.
[83] Dana Pessach and Erez Shmueli. 2022. A review on fairness in machine learning. *ACM Comput. Surveys* 55, 3 (2022), 1–44.
[84] Eric Potash, Joe Brew, Alexander Loewi, Subhabrata Majumdar, Andrew Reece, Joe Walsh, Eric Rozier, Emile Jorgenson, Raed Mansour, and Rayid Ghani. 2015. Predictive modeling for public health: Preventing childhood lead poisoning. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2039–2047.
[85] Aahlad Puli, Lily H. Zhang, Eric K. Oermann, and Rajesh Ranganath. 2021. Out-of-distribution generalization in the presence of nuisance-induced spurious correlations. Retrieved from https://arXiv:2107.00520
[86] Sanjay Purushotham, Chuizheng Meng, Zhengping Che, and Yan Liu. 2018. Benchmarking deep learning models on large healthcare datasets. *J. Biomed. Info.* 83 (2018), 112–134.
[87] Novi Quadrianto and Viktoriia Sharmanska. 2017. Recycling privileged learning and distribution matching for fairness. In *Proceedings of the Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems*, Isabelle Guyon, Ulrike von Luxburg, Samy Bengio, Hanna M. Wallach, Rob Fergus, S. V. N. Vishwanathan, and Roman Garnett (Eds.). 677–688.

[88] Novi Quadrianto, Viktoriia Sharmanska, and Oliver Thomas. 2019. Discovering fair representations in the data domain. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR'19)*. Computer Vision Foundation/IEEE, 8227–8236.
[89] John Rawls. 2001. *Justice as Fairness: A Restatement*. Harvard University Press, Cambridge, MA.
[90] John Rawls. 2009. *A Theory of Justice*. Harvard University Press, Cambridge, MA.
[91] Ashkan Rezaei, Rizal Fathony, Omid Memarrast, and Brian Ziebart. 2020. Fairness for robust log loss classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34. 5511–5518.
[92] Ashkan Rezaei, Anqi Liu, Omid Memarrast, and Brian D. Ziebart. 2021. Robust fairness under covariate shift. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 9419–9427.
[93] Yuji Roh, Kangwook Lee, Steven Whang, and Changho Suh. 2020. FR-Train: A mutual information-based approach to fair and robust training. In *Proceedings of the International Conference on Machine Learning*. PMLR, 8147–8157.
[94] Pedro Saleiro, Benedict Kuester, Loren Hinkson, Jesse London, Abby Stevens, Ari Anisfeld, Kit T. Rodolfa, and Rayid Ghani. 2018. Aequitas: A bias and fairness audit toolkit. Retrieved from https://arXiv:1811.05577
[95] Jessica Schrouff, Natalie Harris, Sanmi Koyejo, Ibrahim M. Alabdulmohsin, Eva Schnider, Krista Opsahl-Ong, Alexander Brown, Subhrajit Roy, Diana Mincu, Christina Chen et al. 2022. Diagnosing failures of fairness transfer across distribution shift in real-world medical settings. *Adv. Neural Info. Process. Syst.* 35 (2022), 19304–19318.
[96] Candice Schumann, Xuezhi Wang, Alex Beutel, Jilin Chen, Hai Qian, and Ed H. Chi. 2019. Transfer of machine learning fairness across domains. Retrieved from https://arXiv:1906.09688
[97] Marco Scutari. 2023. fairml: A statistician's take on fair machine learning modelling. Retrieved from https://arXiv:2305.02009
[98] Laleh Seyyed-Kalantari, Guanxiong Liu, Matthew McDermott, Irene Y. Chen, and Marzyeh Ghassemi. 2020. CheXclusion: Fairness gaps in deep chest X-ray classifiers. In *Proceedings of the Pacific Symposium on Biocomputing*. World Scientific, Singapore, 232–243.
[99] Shreya Shankar, Yoni Halpern, Eric Breck, James Atwood, Jimbo Wilson, and D. Sculley. 2017. No classification without representation: Assessing geodiversity issues in open data sets for the developing world. In *Proceedings of the NIPS Workshop on Machine Learning for the Developing World*.
[100] Yuxin Shi, Han Yu, and Cyril Leung. 2023. Towards fairness-aware federated learning. *IEEE Transactions on Neural Networks and Learning Systems* (2023), 1–17. DOI: https://doi.org/10.1109/tnnls.2023.3263594
[101] Harvineet Singh, Rina Singh, Vishwali Mhasawade, and Rumi Chunara. 2019. Fair predictors under distribution shift. In *Proceedings of the NeurIPS Workshop on Fair ML for Health*. Curran Associates.
[102] Harvineet Singh, Rina Singh, Vishwali Mhasawade, and Rumi Chunara. 2021. Fairness violations and mitigation under covariate shift. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency*. ACM, New York, NY, 3–13.
[103] Dylan Slack, Sorelle A. Friedler, and Emile Givental. 2020. Fairness warnings and Fair-MAML: Learning fairly with minimal data. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. ACM, New York, NY, 200–209.
[104] Kihyuk Sohn, David Berthelot, Nicholas Carlini, Zizhao Zhang, Han Zhang, Colin A. Raffel, Ekin Dogus Cubuk, Alexey Kurakin, and Chun-Liang Li. 2020. Fixmatch: Simplifying semi-supervised learning with consistency and confidence. *Adv. Neural Info. Process. Syst.* 33 (2020), 596–608.
[105] Bahar Taskesen, Viet Anh Nguyen, Daniel Kuhn, and Jose Blanchet. 2020. A distributionally robust approach to fair classification. Retrieved from https://arXiv:2007.09530
[106] Philip S. Thomas, Bruno Castro da Silva, Andrew G. Barto, Stephen Giguere, Yuriy Brun, and Emma Brunskill. 2019. Preventing undesirable behavior of intelligent machines. *Science* 366, 6468 (2019), 999–1004.
[107] Berk Ustun, Yang Liu, and David Parkes. 2019. Fairness without harm: Decoupled classifiers with preference guarantees. In *Proceedings of the International Conference on Machine Learning*. PMLR, 6373–6382.
[108] Victor Veitch, Alexander D'Amour, Steve Yadlowsky, and Jacob Eisenstein. 2021. Counterfactual invariance to spurious correlations in text classification. *Adv. Neural Info. Process. Syst.* 34 (2021), 16196–16208.
[109] Sahil Verma and Julia Rubin. 2018. Fairness definitions explained. In *Proceedings of the IEEE/ACM International Workshop on Software Fairness (FAIRWARE'18)*. IEEE, 1–7.
[110] Sandra Wachter, Brent Mittelstadt, and Chris Russell. 2021. Why fairness cannot be automated: Bridging the gap between EU non-discrimination law and AI. *Comput. Law Secur. Rev.* 41 (2021), 105567.
[111] Haotao Wang, Junyuan Hong, Jiayu Zhou, and Zhangyang Wang. 2023. How robust is your fairness? Evaluating and sustaining fairness under unseen distribution shifts. *Trans. Mach. Learn. Res.* (2023). Retrieved from https://openreview.net/forum?id=11pGlecTz2
[112] Jialu Wang, Yang Liu, and Caleb Levy. 2021. Fair classification with group-dependent label noise. In *Proceedings of the ACM Conference on Fairness, Accountability, and Transparency*. ACM, New York, NY, 526–536.
[113] Serena Wang, Wenshuo Guo, Harikrishna Narasimhan, Andrew Cotter, Maya Gupta, and Michael Jordan. 2020. Robust optimization for fairness with noisy protected groups. *Adv. Neural Info. Process. Syst.* 33 (2020), 5190–5203.

[114] Shirly Wang, Matthew B. A. McDermott, Geeticka Chauhan, Marzyeh Ghassemi, Michael C. Hughes, and Tristan Naumann. 2020. MIMIC-extract: A data extraction, preprocessing, and representation pipeline for MIMIC-III. In *Proceedings of the ACM Conference on Health, Inference, and Learning*. ACM, New York, NY, 222–235.

[115] Hilde Weerts, Miroslav Dudík, Richard Edgar, Adrin Jalali, Roman Lutz, and Michael Madaio. 2023. Fairlearn: Assessing and improving fairness of AI systems. Retrieved from https://arXiv:2303.16626

[116] Jenna Wiens, John Guttag, and Eric Horvitz. 2014. A study in transfer learning: Leveraging data from multiple hospitals to enhance hospital-specific predictions. *J. Amer. Med. Info. Assoc.* 21, 4 (2014), 699–706.

[117] Jakub Wiśniewski and Przemysław Biecek. 2021. fairmodels: A flexible tool for bias detection, visualization, and mitigation. Retrieved from https://arXiv:2104.00507

[118] Blake Woodworth, Suriya Gunasekar, Mesrob I. Ohannessian, and Nathan Srebro. 2017. Learning non-discriminatory predictors. In *Proceedings of the Conference on Learning Theory*. PMLR, 1920–1953.

[119] Taeho Yoon, Jaewook Lee, and Woojin Lee. 2020. Joint transfer of model knowledge and fairness over domains using wasserstein distance. *IEEE Access* 8 (2020), 123783–123798.

[120] Mikhail Yurochkin, Amanda Bower, and Yuekai Sun. 2020. Training individually fair ML models with sensitive subspace robustness. In *Proceedings of the International Conference on Learning Representations*.

[121] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez Rodriguez, and Krishna P. Gummadi. 2017. Fairness beyond disparate treatment and disparate impact: Learning classification without disparate mistreatment. In *Proceedings of the 26th International Conference on World Wide Web*. 1171–1180.

[122] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez Rogriguez, and Krishna P. Gummadi. 2017. Fairness constraints: Mechanisms for fair classification. In *Proceedings of the International Conference on Artificial Intelligence and Statistics*. PMLR, 962–970.

[123] John R. Zech, Marcus A. Badgeley, Manway Liu, Anthony B. Costa, Joseph J. Titano, and Eric Karl Oermann. 2018. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLoS Med.* 15, 11 (2018), e1002683.

[124] Rich Zemel, Yu Wu, Kevin Swersky, Toni Pitassi, and Cynthia Dwork. 2013. Learning fair representations. In *Proceedings of the International Conference on Machine Learning*. PMLR, 325–333.

[125] Wenbin Zhang and Albert Bifet. 2020. Feat: A fairness-enhancing and concept-adapting decision tree classifier. In *Proceedings of the International Conference on Discovery Science*. Springer, 175–189.

[126] Wenbin Zhang, Albert Bifet, Xiangliang Zhang, Jeremy C. Weiss, and Wolfgang Nejdl. 2021. FARF: A fair and adaptive random forests classifier. In *Proceedings of the Advances in Knowledge Discovery and Data Mining: 25th Pacific-Asia Conference (PAKDD'21)*. Springer, 245–256.

[127] Wenbin Zhang, Mingli Zhang, Ji Zhang, Zhen Liu, Zhiyuan Chen, Jianwu Wang, Edward Raff, and Enza Messina. 2020. Flexible and adaptive fairness-aware learning in non-stationary data streams. In *Proceedings of the IEEE 32nd International Conference on Tools with Artificial Intelligence (ICTAI'20)*. IEEE, 399–406.

[128] Xueru Zhang, Ruibo Tu, Yang Liu, Mingyan Liu, Hedvig Kjellstrom, Kun Zhang, and Cheng Zhang. 2020. How do fair decisions fare in long-term qualification? *Adv. Neural Info. Process. Syst.* 33 (2020), 18457–18469.

[129] Yi Zhang, Mengjia Wu, George Yijun Tian, Guangquan Zhang, and Jie Lu. 2021. Ethics and privacy of artificial intelligence: Understandings from bibliometrics. *Knowl.-Based Syst.* 222 (2021), 106994.

[130] Chen Zhao and Feng Chen. 2019. Rank-based multi-task learning for fair regression. In *Proceedings of the IEEE International Conference on Data Mining (ICDM'19)*. IEEE, 916–925.

[131] Chen Zhao and Feng Chen. 2020. Unfairness discovery and prevention for few-shot regression. In *Proceedings of the IEEE International Conference on Knowledge Graph (ICKG'20)*. IEEE, 137–144.

[132] Chen Zhao, Feng Chen, and Bhavani Thuraisingham. 2021. Fairness-aware online meta-learning. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, New York, NY, 2294–2304.

[133] Chen Zhao, Feng Chen, Zhuoyi Wang, and Latifur Khan. 2020. A primal-dual subgradient approach for fair meta learning. In *Proceedings of the IEEE International Conference on Data Mining (ICDM'20)*. IEEE, 821–830.

[134] Chen Zhao, Changbin Li, Jincheng Li, and Feng Chen. 2020. Fair meta-learning for few-shot classification. In *Proceedings of the IEEE International Conference on Knowledge Graph (ICKG'20)*. IEEE, 275–282.

[135] Jieyu Zhao, Yichao Zhou, Zeyu Li, Wei Wang, and Kai-Wei Chang Chang. 2018. Learning gender-neutral word embeddings. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*.

[136] Zirui Zhou, Lingyang Chu, Changxin Liu, Lanjun Wang, Jian Pei, and Yong Zhang. 2021. Towards fair federated learning. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, New York, NY, 4100–4101.

[137] Fuzhen Zhuang, Zhiyuan Qi, Keyu Duan, Dongbo Xi, Yongchun Zhu, Hengshu Zhu, Hui Xiong, and Qing He. 2020. A comprehensive survey on transfer learning. *Proc. IEEE* 109, 1 (2020), 43–76.

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